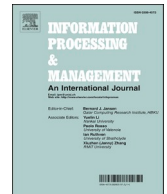




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Official or Influencer? An AI-Enhanced Analytical Framework for Decoding Multimodal Persuasion in Government Marketing Videos

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ABSTRACT

Local officials increasingly use short-video platforms for direct-to-consumer marketing, creating tension between bureaucratic identity and platform logic. This multimodal persuasion context is difficult to analyze at scale without losing theoretical depth. To address this gap, we propose and validate an AI-enhanced analytical framework centered on a "Human-AI Dynamic Cognitive Dialogue" protocol, which turns the Multimodal Large Language Model (MM-LLM) from an automated "black box" into a theory-guided analytical mechanism and reduces the text-centric bias of traditional computational methods. Crucially, it uses theory-driven human oversight to guide AI-assisted induction, preserving methodological transparency while eliciting strategy configurations beyond researcher priors. Based on 779 Douyin videos from Chinese officials, the study identifies a "bifurcation of authority": authority cues are differentially associated with engagement depending on personified presentation (e.g., official attire) rather than institutionalized context (e.g., a government office). Engagement-cost differentiation is also evident: higher-cost actions such as sharing correspond to a synergy of personified authority, narrative resonance, and high cognitive value. Smaller accounts show a "legitimacy compensation" pattern, with strong authority signals associated with higher engagement among accounts lacking established social capital. The study establishes a replicable protocol for decoding unstructured video data and offers a bounded extension of the Elaboration Likelihood Model by showing how dynamic multimodal cues correspond to differentiated engagement patterns in algorithmic short-video environments.

1. Introduction

Amid the global tide of digitalization, the interaction between governments and the public is undergoing a profound shift toward "personalization," as officials increasingly act as charismatic communication agents (Bangero, 2025) who engage citizens directly via social media (e.g., Ecker-Ehrhardt, 2023; Özdemir et al., 2025). In China's distinctive governance context, grassroots officials have extended this trend by using short videos for direct-to-consumer (D2C) marketing, thereby making it a more common practice (Cheng et al., 2024). They must continually negotiate between the "bureaucratic logic" that sustains institutional authority and the "influencer logic" that caters to platform algorithms and high engagement (Chen & Gong, 2025). This complex identity negotiation is rendered

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entirely through the multimodal signals of the short-video medium.

However, investigating this phenomenon presents a dual challenge that intertwines theoretical ambiguity with methodological limitations. Theoretically, while the Elaboration Likelihood Model (ELM) explains persuasion via central and peripheral routes, it was originally conceptualized for static media (Chandrasekaran et al., 2025). It remains unclear how conflicting cues, such as the "bureaucratic" need for authority and the "influencer" need for authenticity, interact dynamically within a short video to shape user engagement (Lu & Shen, 2023). Specifically, existing literature has not sufficiently explored how these multimodal strategies are differentially associated with behaviors with varying cognitive costs (e.g., the low cost of a "like" versus the high social cost of a "share") (Song et al., 2026). Without addressing this, current theoretical frameworks may struggle to fully account for the temporal nuances of persuasion specific to algorithm-driven streaming media environments.

Methodologically, bridging this theoretical gap is hindered by the intrinsic complexity of unstructured video data (Yin et al., 2025). Traditional content analysis struggles to capture the dynamic interplay of visual and auditory cues at scale (Kirilenko & Stepchenkova, 2025), while rule-based computational pipelines often operate in a "semantic vacuum" lacking theoretical guidance (Tong et al., 2022). Furthermore, although Multimodal Large Language Models (MM-LLM) offer promise, deploying them as unsupervised "black boxes" introduces systematic biases and epistemic uncertainties (e.g., Hu et al., 2023; Qi et al., 2023). Consequently, scholars face a methodological bottleneck. Without a replicable protocol to analyze video data at scale, it remains difficult to empirically examine how conflicting identity cues interact over time and relate to user behavior.

To address these challenges and move beyond treating AI as an automated "black box," this study develops an AI-enhanced multimodal content analysis framework that positions LLMs as analytic mechanisms within structured human-AI cognitive collaboration. Guided by human theoretical judgment, the framework counteracts modality bias, supports rigorous qualitative procedures (Gao et al., 2024), and uses an iterative "dynamic cognitive dialogue" protocol to induce and validate communication strategies from video data. Its value lies in combining scale with human-AI complementarity, allowing emergent strategy configurations beyond researchers' priors and supporting deeper theoretical insight than either agent could achieve alone (Kirilenko & Stepchenkova, 2025; Lu & Zhang, 2024; Xu et al., 2026).

To systematically bridge the theoretical gaps identified above using this methodological innovation, this study pursues three specific research objectives: (1) To construct a theoretically grounded, human-in-the-loop AI framework for decoding multimodal government communication; (2) To empirically identify the strategy configurations used by officials to navigate the tension between bureaucratic and platform logics; (3) To examine how these strategies are differentially associated with public engagement behaviors across varying levels of cognitive cost. Guided by these objectives, we propose the following research questions:

- RQ1: What key multimodal combinations of strategies do grassroots officials employ in D2C marketing videos to balance their dual identities as "government officials" and "influencers"?
- RQ2: How are combinations of strategies differentially associated with public engagement behaviors across varying cost levels (likes, comments, shares)?
- RQ3: To what extent can the AI-enhanced analytical framework reliably and validly parse complex video content?

To achieve these objectives, this study grounds its analysis in a dataset of 779 videos from Chinese officials on Douyin, utilizing a validated "Human-AI Dynamic Cognitive Dialogue" protocol to extract high-dimensional semantic features. Our empirical results document a "bifurcation of authority," where personified cues are positively associated with engagement while institutional contexts are negatively associated with it, and support the interpretation of distinct "cost ladders" for digital persuasion. By establishing this replicable methodological blueprint, the study contributes to Information Management by offering a valid protocol for unstructured video analysis, and extends the Elaboration Likelihood Model into the algorithmic era by elucidating the dynamic interplay of conflicting persuasion cues.

2. Theoretical Background

2.1. Identity Negotiation Dilemmas in Platformized Governance

When grassroots officials enter digital platforms as content creators, they are immediately situated between two fundamentally conflicting institutional logics: a traditional bureaucratic logic and an emergent platform logic. Bureaucratic logic, rooted in hierarchy and procedural compliance, prioritizes authoritative and orderly information transmission; platform logic, governed by algorithmic visibility mechanisms, rewards personalized and affectively charged content that elicits high engagement. This institutional complexity compels grassroots officials to engage in continuous identity work, seeking a dynamic equilibrium between maintaining the legitimacy of their bureaucratic role and meeting the platform's expectation of "authenticity" (J. Kim & Kim, 2024). Strategically, they cultivate a hybrid identity that preserves the authority of an official representative while exhibiting personal qualities that can evoke emotional resonance.

Existing scholarship has notable blind spots in explaining this phenomenon. Research on government social media has examined policy communication at the macro level (Qin et al., 2025) and governance objectives (Guo et al., 2025), yet it often treats individual officials as passive executors of institutional will, overlooking their agentic, micro-level strategic choices. Meanwhile, scholarship on digital platforms has analyzed how algorithms shape content production (Chen & Shi, 2025) but rarely considers the distinctive dynamics that arise when communicators wield public authority. More critically, the literature has not established a coherent analytical chain from the macro-sociological question of why officials must negotiate identity (Chen & Gong, 2025) to the

micro-communication question of how they accomplish this negotiation through concrete information strategies (Grantham et al., 2025). To bridge this gap, we first turn to established persuasion theory to map the mechanisms available to these officials.

2.2. The Elaboration Likelihood Model in Digital Persuasion

To systematically decode the specific strategies officials employ, we adopt the Elaboration Likelihood Model (ELM) as our foundational theoretical lens. As a dominant framework in computer-mediated communication, ELM posits that persuasion occurs through two distinct cognitive pathways based on the audience's motivation and ability to process information (Petty & Cacioppo, 1986).

The central route involves high cognitive elaboration, where audiences critically evaluate the quality of arguments and the substance of the message. In this study, a construct is treated as central-route when its persuasive relevance depends primarily on audiences' processing of argument quality, informational density, message substance, or cognitive utility. In the context of government D2C marketing, this route corresponds not only to the strength of rational policy appeals but also to the density of high-utility product information (Wang et al., 2025). Both dimensions represent substantive cognitive value, whether for institutional legitimacy or transactional utility, and require sustained viewer attention and cognitive effort to process. Conversely, the peripheral route operates when cognitive elaboration is low. A construct is treated as peripheral-route when its persuasive relevance is expected to operate mainly through source heuristics, visual authority, affective resonance, scene context, editing rhythm, or directive prompts that can be evaluated with limited elaboration. Here, audiences rely on heuristic cues, such as official uniforms, visual aesthetics, or emotional resonance, to make quick judgments without scrutinizing the underlying arguments (Xu et al., 2025). Ideally, this dual-process model allows officials to balance their strategies: using central cues to convey bureaucratic legitimacy and peripheral cues to meet platform demands for engagement. However, while ELM provides a structural framework for classifying these cues, it does not fully account for the relational dynamics inherent to social media. Consequently, we extend this theoretical lens by incorporating social identity, perceived social presence, and trust formation as explanatory mechanisms (Li et al., 2024). This integration clarifies why specific authoritative or affective cues succeed or fail within ELM's cognitive routes depending on the official's evolving relationship with the audience. These criteria provide the conceptual basis for the construct mapping used in the empirical analysis.

2.3. The Temporal Challenge: Reassessing Persuasion in Dynamic Media

However, applying this classic framework to the attention-scarce environment of short videos presents fundamental theoretical challenges. The Limited Capacity Model of Motivated Mediated Message Processing (LC4MP) suggests that before any ELM-based processing can occur, content must first survive an initial attentional filter. Specific low-level audiovisual features—such as rapid cutting, color contrast, or abrupt sound changes—trigger orienting responses that forcibly allocate cognitive resources (Lang, 2000). Operating largely outside conscious control, these perceptual cues modulate visual search efficiency and directly shape initial information encoding (Cardamone et al., 2025; Clayton & Myrick, 2025).

Consequently, a digital paradox emerges. The perceptual cues that capture immediate attention often map onto peripheral-route cues in ELM. This creates a potential conflict with officials' intention to communicate authoritative, reasoned messages via the central route. If officials over-rely on "influencer-style" peripheral cues to clear the attentional hurdle, they risk eroding the central-route authority required for bureaucratic legitimacy.

Yet, because the orienting response is transient and fleeting (Potter & Bolls, 2012), static variable coding fails to capture how attention is sustained or lost along the video timeline. This reveals a deeper challenge: classic persuasion theories were essentially developed for static information (e.g., print ads or text posts) and thus face limitations when applied to short video, a medium characterized by temporal dependency. The persuasive force of short video may not arise from a static bundle of multimodal cues alone, but from their temporal evolution and cumulative patterns (Sim et al., 2024). For instance, a cut from a government office to an agricultural field derives its persuasive impact from the dynamic shift itself, which cannot be reconstructed by simply summing separate static codings of the two scenes (Wu et al., 2025). This dynamic complexity exposes an internal deficit in existing theory: current frameworks lack the constructs needed to describe and explain how persuasive effects accumulate and transform over time.

Temporal dependency is therefore not merely a methodological obstacle but a profound theoretical challenge—current persuasion theories remain ill-equipped to fully account for information processing in dynamic media environments. To resolve these theoretical frictions, our AI-enhanced framework shifts the unit of interpretation from isolated static feature sets to temporally ordered multimodal cue configurations. This refinement does not reject ELM's dual-route logic; rather, it specifies how central and peripheral cues unfold, combine, and change salience across short-video sequences. This shift allows us to analyze more directly how multimodal strategies evolve and interact over time, rather than viewing them as static signals.

3. Data

To test the framework, our data strategy constructed an empirical corpus reflecting direct-to-consumer (D2C) marketing by "government official influencers" in a complex digital media environment. We selected Douyin, China's leading short-video platform, because its algorithmically driven distribution and large user base provide a natural setting for observing digital engagement behaviors (Chen & Shi, 2025). This setting is analytically appropriate but platform-specific: Douyin's recommendation logic, creator norms, and short-video culture may shape which official videos become visible and how users interact with them. Against the policy backdrop of rural revitalization, many grassroots officials use the platform personally to promote local economies, creating naturally occurring data on the negotiation between bureaucratic and influencer identities. Our multi-stage, theory-driven sampling process (Fig. 1)

supports the analytic validity and consistency of the final sample within this defined platform context.

Corpus construction began with purposive sampling to identify and isolate official accounts aligned with our research constructs. We adopted three stringent inclusion criteria: first, the account had to bear an official institutional verification badge or be verifiably affiliated with public office through open sources; second, its content had to focus on substantive promotion of local economic resources rather than purely policy publicity or personal life-logging; third, the account needed to remain active throughout the observation window (from June to August 2025). Using a combination of algorithmic recommendations and systematic keyword search, we initially identified 159 qualifying accounts. To extract content highly relevant to D2C marketing from the tens of thousands of videos posted by these accounts, we implemented a capped extraction strategy: for each account, we retrieved up to the top 20 videos ranked by public engagement (sum of likes, comments, and shares). For accounts with a total volume below this threshold, the full available corpus was retained to ensure comprehensive coverage, resulting in an initial pool of 3,149 videos (with an average of 19.8 videos per account due to the capped extraction strategy and varying account activity levels). This strategy prioritizes analytically relevant and visible D2C communication, but it may also overrepresent content that had already attracted engagement or algorithmic amplification and underrepresent low-exposure or less discoverable videos. We then addressed the challenge of detecting marketing intent by implementing a hierarchical, lexicon-based weighted scoring heuristic to filter textual metadata (descriptions and hashtags). This deterministic strategy assigned cumulative relevance scores based on three positive tiers, specific commodity entities (weight +3), explicit transactional markers (e.g., "buy," "link"; weight +2), and general rural revitalization contexts (weight +1), while penalizing unrelated governance topics (weight -2) and provisionally retaining entries with insufficient information to minimize false negatives, reducing the candidate set to 847. To validate algorithmic screening, two trained research assistants independently verified this subset, achieving excellent inter-coder reliability (Krippendorff's $\alpha = 0.947$), with disagreements adjudicated by a third coder. Finally, to bound computational scale and define analytic scope, we excluded videos longer than 300 seconds. The resulting analytic sample comprised 779 valid videos spanning 23 provincial-level regions across China, providing a robust empirical foundation for the subsequent analyses within the stated sampling scope.

For multimodal analysis, we performed systematic feature engineering for each video in the final sample. First, all audio tracks were extracted and standardized to facilitate automatic speech transcription and acoustic feature extraction. On the visual side, to balance the capture of key narrative shifts with the computational cost of AI analysis, we implemented a duration-adaptive keyframe sampling strategy. Rather than uniform or random sampling, this strategy systematically targets key visual information near the opening, middle, and closing segments, thereby preserving core temporal cues while optimizing efficiency—an especially important consideration for dynamic video data. Let T denote the total video duration (in seconds). The set of keyframe timestamps $S(T)$ is defined as:

$$\mathcal{S}(T) = \{2, 5, 10, 15\} \cup \{t \in \{30, 45, 60, 90, 120, 150, 180\} | t < T\}$$

$$\cup \{45k | 225 \leq 45k < T\} \cup \{T - 1\}$$

In parallel, we collected platform-visible engagement metrics for each video (counts of likes, comments, and shares), metadata

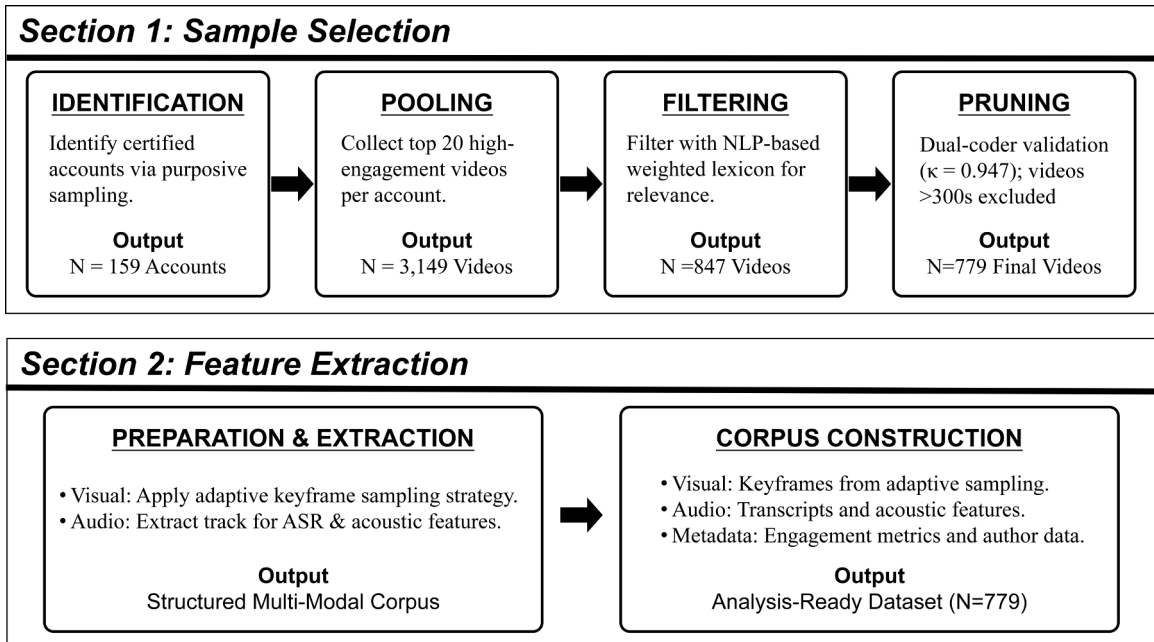


Fig. 1. Data Collection and Sampling Pipeline. Note: The filtration process reduced 159 qualifying accounts to 779 videos through purposive sampling, engagement-based pooling, and NLP-based filtering.

(titles and tags), and publisher information. These measures capture observable engagement behaviors on Douyin rather than direct indicators of complete public attitudes, persuasion outcomes, or offline behavioral change. We also cross-validated and compiled background information for each official— such as region and administrative rank; these characteristics were included as control variables in the empirical models. Taken together, these procedures yielded a high-dimensional, analysis-ready multimodal dataset that provides a reliable empirical basis for implementing our AI-enhanced framework and examining associations between multimodal strategies and observable platform engagement.

4. Methodology: An AI-Enhanced Multimodal Content Analysis Framework

To analyze grassroots officials' multimodal persuasion strategies and their associations with public engagement, we developed a five-stage AI-enhanced analytical framework. Fig. 2 shows how the framework integrates theory-driven deduction and data-driven induction through structured human-AI collaboration (see Appendix Table A1) to extract and validate strategies from 779 videos. For procedural transparency, the analytical chain is: raw videos → adaptive keyframes and audio transcripts → MM-LLM semantic features → BERTopic topics → HAC-supported strategy constructs → ELM theory anchoring → validated coding rules → full-corpus variables → empirical models and reported findings. Each stage specifies its input, AI task, human oversight, output, and validation function in the text or appendix, treating AI assistance as an auditable workflow rather than an automated decision process.

4.1. From Visual Signals to Semantic Features: Dynamic Cognitive Dialogue

To capture the temporal dependency overlooked by conventional static-frame analysis, the first stage implements a dynamic cognitive dialogue protocol designed to induce features directly from the data. From a theory-driven sample of 78 high-engagement videos, we extracted keyframe sequences using our adaptive strategy.

As illustrated in Fig. 3, we use Gemini 2.5 Pro as the multimodal inference engine within a sequential context-injection workflow to process sampled frames together with the audio track. We selected this model for its strong multimodal performance and long-context capability, which enables the same analytical episode to retain an audio-derived semantic baseline while moving through the keyframe sequence (Comanici et al., 2025). This design minimizes information loss that can arise when modalities are encoded separately. Operationally, the model first used the transcript/audio and metadata to establish the product category, communicative setting, and emotional tone; it then interpreted each keyframe against this baseline and the accumulated prior-frame context. Crucially, to capture temporal evolution, the model performs a dynamic contrastive analysis: for each frame at time *t*, it compares the new visual evidence against the immediately preceding frame at time *t*–1, whose analysis is provided as context (see Appendix Table A2). The comparison focused on newly appearing, disappearing, intensified, or weakened cues relevant to the coding rules, rather than treating each frame as an isolated image. This procedure captures meaning as it unfolds over time and yields 4,680 structured features. Each visual event—whether a stable attribute or a change—was annotated with its inferred communicative function, providing a fine-grained semantic foundation for subsequent analysis.

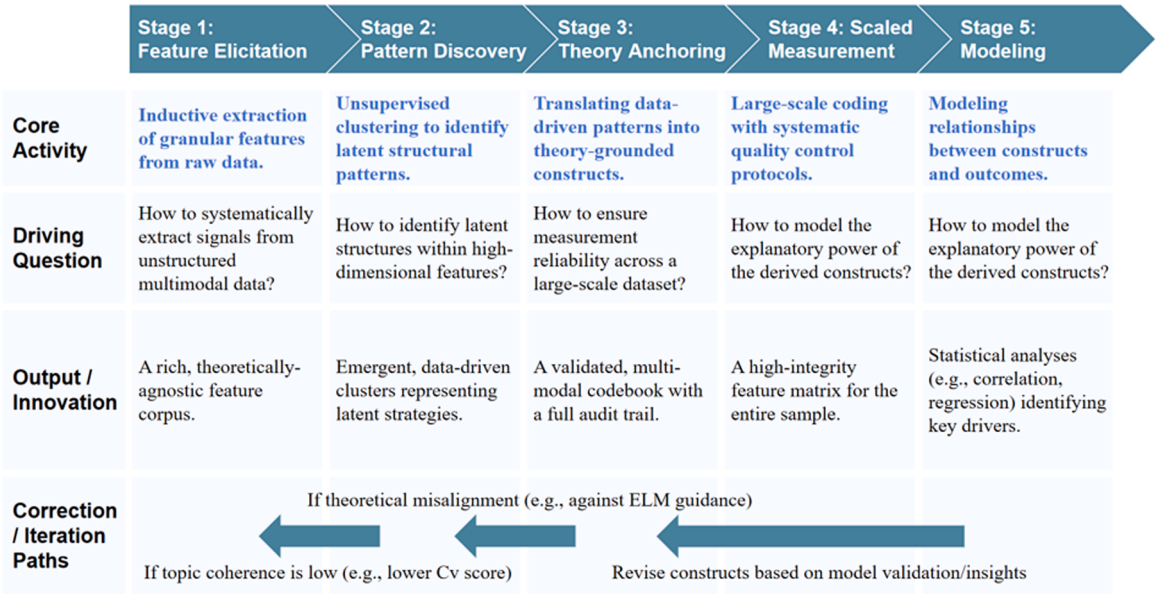


Fig. 2. The Five-Stage AI-Enhanced Analytical Framework for Multimodal Content. Note: The workflow integrates inductive feature elicitation with deductive measurement. Each stage specifies the analytical input, AI operation, human review function, output, and validation or correction path.

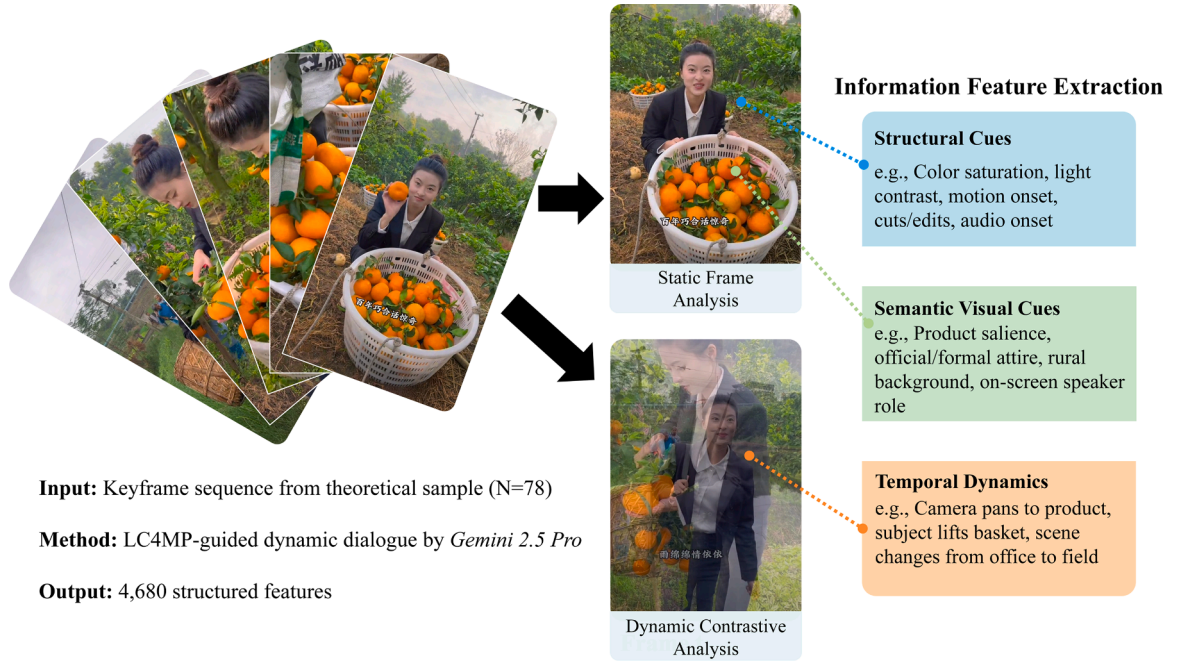


Fig. 3. The AI-Enhanced Feature Extraction Process via Dynamic Contrastive Analysis.

4.2. From Semantic Features to Strategy Combinations: Unsupervised Pattern Discovery

This stage aims to convert unstructured visual signals into interpretable strategy categories without imposing theoretical biases. The input to this stage was not the raw video itself but the set of MM-LLM-generated semantic annotations produced in Stage 1, which are short-text-like, semantically dense descriptions of visual, audio, and temporal cues. To minimize researcher priors and efficiently identify strategy combinations from large volumes of atomic features, the second stage adopts a data-driven inductive procedure. We selected BERTopic over traditional probabilistic frameworks, such as Latent Dirichlet Allocation (LDA), specifically to mitigate the data sparsity inherent in short-text analysis (Benz et al., 2025). This choice is a methodological fit argument rather than a claim of universal superiority: bag-of-words topic models are less suitable when concise annotations contain semantically related but lexically diverse descriptions, whereas BERTopic's contextual embeddings are better aligned with the study's intermediate data structure. Unlike the bag-of-words assumptions of LDA, BERTopic leverages S-BERT contextual embeddings to capture deep semantic dependencies, ensuring robust topic representation even within concise visual annotations.

Model quality was evaluated using C_v topic coherence. The algorithm initially identified 85 topics with an average coherence of 0.504, indicating meaningful latent structure. This score was used as a diagnostic indicator of latent semantic structure rather than as the sole criterion for accepting or rejecting topics. To rigorously refine these clusters, we implemented a dual-filter protocol (Son & Park, 2025): Quantitatively, we excluded outlier topics that failed to meet the minimum document frequency threshold; qualitatively, we manually pruned "noise topics" stemming from OCR errors or irrelevant background descriptors (e.g., generic "sky") that lacked communicative intent. Topic refinement therefore combined quantitative topic size, representative documents, semantic interpretability, and researcher review of communicative relevance, rather than relying mechanically on the coherence score alone.

The remaining 47 validated topics were then synthesized into nine preliminary strategic constructs via Hierarchical Agglomerative Clustering (HAC). For this step, the clustering unit was the validated topic rather than the individual video or raw semantic annotation. Each topic was represented as a TF-IDF topic-document vector derived from its representative topic documents. We employed Ward's linkage with Euclidean distance to maximize within-cluster homogeneity, effectively forming compact higher-order candidate groups from semantically proximate topics. The hierarchical tree was cut to yield nine candidate clusters, with the corresponding HAC dendrogram reported in Appendix Fig. A1. This solution was retained because it provided the clearest balance among semantic coherence, non-redundancy, interpretability, and fit with the ELM-oriented strategy distinctions used in the following theory-anchoring stage. HAC served as an organizing device for grouping semantically proximate topics; final construct boundaries were further audited through c-TF-IDF keyword inspection, representative examples, and preparatory semantic review before the formal ELM theory anchoring described later. Finally, to substantiate the semantic validity of this aggregation, we inspected the top class-based TF-IDF (c-TF-IDF) keywords associated with representative topics. For instance, the construct "Source Official Identity Salience" was anchored by high-weight terms such as "uniform," "secretary," and "inspection," whereas "Engaging Narrative Style" emerged from clusters defined by "plot twist," "funny," and "contrast plot," confirming that the derived constructs possess distinct and theoretically coherent boundaries.

4.3. From Strategy Combinations to Theoretical Constructs: Human-AI Collaborative Theory Anchoring

To systematically connect data-driven discoveries to established theory, the third stage uses a guided theory-building protocol that converts preliminary strategy constructs into measurement tools testable within the Elaboration Likelihood Model (ELM). The protocol guides the large language model through three hierarchical steps: first, empirical deconstruction of each preliminary construct to identify its core elements; second, distillation of the underlying strategy theme and hypothesis generation regarding its psychological mechanism; and third, mapping the strategy to ELM's central or peripheral route and producing operational definitions and measurement scales. The route assignment followed the criteria: central-route constructs were those whose persuasive relevance depended on argument quality, informational density, message substance, or cognitive utility, whereas peripheral-route constructs were those expected to operate primarily through source heuristics, visual authority, affective resonance, scene context, editing rhythm, or directive prompts. This mapping was completed during the theory-anchoring stage, before scaled coding of the full corpus and before statistical modeling; it was therefore not derived from the direction, magnitude, or significance of the regression coefficients. This procedure ensures that each transformation—from raw features to final constructs—has explicit decision rationales, establishing a complete audit trail. For example, the formation of Source Official Identity Salience progresses from the atomic feature "uniform present" to the theme of "authoritative visual cues," which is then mapped to the ELM peripheral route via source-credibility mechanisms and operationalized as a binary variable (see Table 1).

4.4. Measurement Validation and Scaled Application

To ensure measurement robustness and realize the benefits of human-AI complementarity, we implemented a systematic validation and application workflow.

4.4.1. Triangulated Validation and Optimization Based on Cognitive Complementarity

Guided by the principle of human-AI cognitive complementarity, we conducted triangulated reliability diagnostics on a 78-video subset (AI-AI, human-human, and AI-human reliability) to delineate the optimal division of labor. Fig. 4 visualizes the inter-rater reliability diagnostics (refer to the Figure Note for interpretation thresholds), revealing that AI exhibited superior internal consistency on structured visual constructs ($\alpha_{AI-AI} > 0.950$). Specifically, the model consistently outperformed human coders in detecting objective patterns such as Source Official Identity Salience and Scene Official Context Salience. This advantage reflects AI's strict adherence to visual patterns; for instance, in judging Product Visual Salience, AI classified 81.78% of cases as "not salient," compared with 75.56% for humans, indicating a more conservative yet highly consistent standard.

By contrast, when tasks required contextual understanding and affective inference, human cognition remained indispensable. Engaging Narrative Style most clearly manifested this difference: human-human reliability ($\alpha_{Human-Human} = 0.825$) far exceeded AI's internal reliability ($\alpha_{AI-AI} = 0.642$), the latter falling below acceptable thresholds. Qualitative diagnostics suggested distinct judgment mechanisms: humans could register tacit cues such as intonation and pausing, identifying narrative engagement in 72.37% of cases versus 67.11% for AI.

These diagnostics informed iterative optimization. For constructs on which AI underperformed, we decomposed abstract concepts into concrete behavioral indicators that AI could reliably detect (e.g., breaking Engaging Narrative Style into first-person narration and intonational variation), raising AI internal reliability above 0.85. In parallel, we established a dedicated audio analysis pipeline, using

Table 1
Preliminary Theoretical Constructs and ELM Mapping.

Construct Name	Object of Assessment	ELM Route & Interpretation
Source Official Identity Salience	Visual appearance of the speaker/uploader	Peripheral route. An official identity is a classic peripheral cue representing source authority and credibility. When audiences are unwilling or unable to engage in deep content processing, this cue can rapidly establish trust.
Scene Official Context Salience	Visual background of the video's focal scene	Peripheral route. As a contextual cue, the scene amplifies the source's official attributes and provides a legitimacy endorsement for the content without requiring further evidence.
Content Rational Appeal Strength	Textual properties of the complete content (audio and script)	Central route. When audiences have sufficient motivation and ability, stronger rational argumentation is perceived as more compelling evidence.
Engaging Narrative Style	Full audio transcript	Peripheral route. Through affective expression, it directly elicits audience liking and resonance, fostering shallow interactions such as likes and shares.
Editing Frequency	Visual editing techniques of the video	Peripheral route. A highly dynamic visual presentation functions as a sensory stimulus that effectively attracts and sustains audience attention.
Call to Action	Complete video content (audio and on-screen text)	Peripheral route. As a direct, low-cognition directive cue, a call to action is intended to bypass elaborate processing.
Product Visual Salience	Visual focal point within the frame	Peripheral route. By making the product the primary visual focus, the video simplifies its persuasive message and reduces the audience's cognitive processing costs.
Content Information Complexity	Audio content	Central route. Higher informational complexity implies greater cognitive resources are required for processing.
Marketing Information Intensity	Audio content	Central route. This variable directly gauges the argument strength of the core message (the product).

Note: ELM route labels were assigned before scaled coding and modeling. Central-route constructs involve argument quality, informational density, message substance, or cognitive utility; peripheral-route constructs rely on source, affective, visual-contextual, temporal, or directive heuristics.

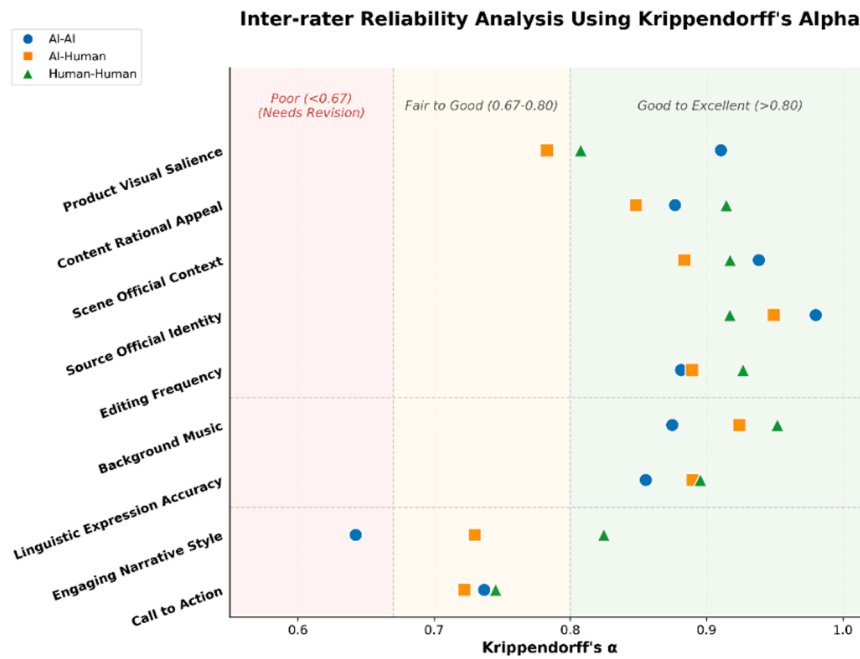


Fig. 4. Inter-rater Reliability Analysis Using Krippendorff's Alpha. Note: Marker distance indicates the AI-human cognitive-complementarity gap; wider gaps on subjective constructs indicate greater need for human adjudication.

Table 2

The Theoretical Construct System of Video Strategies.

Construct Name	Object of Assessment	Quantification Rule Summary	ELM Route & Interpretation
Source Official Identity	Visual appearance of the speaker/uploader	Binary: 1 (salient), the speaker wears a uniform, formal attire, or official insignia; 0 (non-salient), casual or ethnic attire.	Peripheral route. Official identity functions as an authority cue that rapidly fosters trust in low-involvement contexts.
Scene Official Context	Visual background of the video's focal scene	Binary: 1 (formal), government/institutional spaces or displays of official symbols; 0 (informal), everyday life or production settings.	Peripheral route. Contextual cues strengthen the source's official attributes and provide a legitimacy endorsement.
Content Rational Appeal	Textual properties of the complete content (audio + subtitle)	Binary: 1 (rational); primarily conveys objective information; 0 (emotional) — primarily aims at affective arousal.	Central route. The strength of rational argumentation drives deep cognitive processing and trust formation.
Engaging Narrative Style	Full audio transcript	Binary (AI-coded): 1 (present): overtly passionate or emotional expression; 0 (absent) — calm, matter-of-fact narration.	Peripheral route. Affective expression evokes resonance and promotes shallow interactions.
Editing Frequency	Visual editing of the video	Binary: 1 (high dynamism): ≥ 2 scene switches; 0 (low dynamism) — < 2 switches.	Peripheral route. Visual dynamism sustains attention and enhances entertainment value.
Call to Action	Complete video content (audio and on-screen text)	Binary: 1 (present): contains cues that guide specific behaviors; 0 (absent) — no directive cues.	Peripheral route. Direct instructions bypass elaborate processing and prompt targeted behavior.
Product Visual Salience	Visual focal point within the frame	Binary: 1 (salient): the product is the sole visual subject and no human face appears; 0 (non-salient) — otherwise.	Peripheral route. Visual simplification reduces cognitive processing costs and accelerates attention.
Content Information Complexity	Audio content	Continuous: number of unique words $\div$ total words (lexical diversity).	Central route. Higher informational complexity requires more cognitive resources and deeper processing.
Marketing Information Intensity	Audio content	Continuous: count of product-introduction words $\div$ total words (share of product-intro text).	Central route. The density of core product information shapes deliberative decision-making.
Linguistic Expression Accuracy	Quality of speech-to-text transcript	Binary: 1 (problematic): semantically incoherent; 0 (normal) — semantically clear.	Control variable. Used to account for potential bias due to transcription/data quality.

Note: Binary coding is used for visually salient or directive cues whose meaning depends on presence or absence; continuous coding is used for linguistic and informational constructs whose meaning depends on degree, density, or diversity. ELM labels follow the criteria introduced above and summarized in Table 1.

natural language processing to derive richer linguistic features from transcripts. This optimization, grounded in a principled division of cognitive labor, preserves human insight while maximizing AI's strengths in scalable, consistent processing and thus constitutes the core mechanism safeguarding study reliability. Accordingly, the simplified indicators should be understood as theory-anchored operational proxies rather than arbitrary reductions of complex constructs. For visually anchored cues, binary coding captures whether a salient cue is perceptually available to viewers; for linguistically rich constructs, continuous measures preserve meaningful variation in lexical diversity or product-information density. The resulting construct system (Table 2) reflects this strategy: visually anchored constructs use efficient binary coding, whereas linguistically rich constructs employ finer-grained measurement scales.

4.4.2. AI Double-Blind Coding with Human Adjudication

To systematically decode the multimodal corpus, we engineered a "Sequential Context-Injection Protocol" within the AI coding workflow. The input for each coding case consisted of metadata, the complete transcript/audio context, and the sampled keyframe sequence; the AI task was to apply the finalized codebook to this evidence package, while the human role was to design the codebook, inspect reliability diagnostics, and adjudicate divergent or context-dependent cases. In this protocol, the MM-LLM was mandated to ingest and process the complete audio track in the initial conversational turn, implicitly performing automatic speech recognition and acoustic scene analysis to extract semantic keywords and emotional tones (Comanici et al., 2025). This process established a latent semantic schema (e.g., inferring the specific product category or official context) before the model received any visual frames. By leveraging this pre-loaded auditory context as a cognitive anchor, the protocol ensured cross-modal semantic consistency, aligning the interpretation of subsequent visual cues with the established narrative timeline and significantly improving stability for ambiguous features (e.g., reliability for Editing Frequency rose to 0.932). To ensure reproducibility, we enforced a deterministic generation protocol with strict hyperparameter constraints to maximize output stability (Dewan et al., 2026) (see Appendix Table A2 for detailed prompt logic and settings).

Applying this rigorous protocol to the full set of 779 videos, two independent AI instances coded the corpus separately under identical prompts and coding rules and achieved high inter-rater reliability (average = 0.893). Despite this high agreement, a subset of cases (approximately 10.7%), primarily involving context-dependent constructs, exhibited divergence and required human adjudication using optimized coding rules. Adjudication was reserved for these divergent or context-dependent cases: researchers returned to the transcript/audio context, sampled keyframes, and coding rules, and final codes were assigned based on explicit textual, visual, or audiovisual evidence rather than by averaging AI outputs. Crucially, to address the limitation of framework dependency on a specific model (instrumental validity), we conducted a robustness check on a stratified subsample (N=78) using the efficiency-optimized Gemini 2.5 Flash. Applying identical prompts and logic, the results indicated strong convergence with the final dataset (mean agreement = 94.3%; mean κ = 0.861; see Appendix Fig. A2), supporting procedural stability across two Gemini model variants. This consistency indicates that the framework's reliability derives from the structured "Context-Injection" protocol and rigorous construct definitions rather than the specific idiosyncrasies of a single foundation model. Accordingly, the validation should be read as a diagnostic cross-model check, not as evidence that the procedure is independent of all possible foundation models. Through these quality-control mechanisms, we completed feature extraction for all 779 valid samples.

Table 3
Descriptive Statistics of Raw Engagement Outcomes and Explanatory Variables.

Variable	Mean	SD	Min	Max
Likes	5696.619	24590.528	27	378151
Comments	766.225	4313.593	2	77769
Shares	1641.516	16550.657	1	357835
Content Information Complexity	0.578	0.148	0	1
Marketing Information Intensity	0.226	0.159	0	1
Content Rational Appeal Strength	0.553	0.497	0	1
Source Official Identity Salience	0.281	0.448	0	1
Scene Official Context Salience	0.155	0.362	0	1
Engaging Narrative Style	0.754	0.43	0	1
Editing Frequency	0.777	0.417	0	1
Call to Action	0.44	0.497	0	1
Product Visual Salience	0.228	0.412	0	1
Background Music	0.926	0.263	0	1
Speech Rate	3.476	1.225	0.182	9.567
Linguistic Expression Accuracy	0.084	0.277	0	1
Video Duration (In seconds)	3.424	0.675	1.386	5.704
Follower Count (In followers)	10.507	1.24	7.457	15.569
Publisher Age	38.217	6.864	24	56
Publisher Gender	0.172	0.376	0	1
Publisher Province	2.099	0.823	1	3

Note: Likes, Comments, and Shares are platform-visible raw count outcomes used in the negative binomial models. Video Duration and Follower Count are reported on the log scale; binary variables are coded 0/1.

5. Empirical Analysis

The empirical analysis uses the validated variables produced by the preceding workflow: semantic features extracted from key-frames and transcripts were organized into topics, strategy constructs, coding rules, and full-corpus measures, which are summarized in Table 3 and then entered into the regression models. Our analysis of 779 marketing videos produced by grassroots officials describes how the hybrid "official-influencer" identity materializes as concrete communication strategies. Descriptive statistics (Table 3) first quantify these observable patterns. The evidence indicates a marked de-emphasis on visual symbols associated with the traditional bureaucratic identity: only 28.1% of videos display official attire, and 15.5% use officially coded scene backgrounds. In sharp contrast, creators broadly adopt platform-native elements: 75.4% of videos employ an Engaging Narrative Style, and 77.7% use high Editing Frequency. This strategic pivot from authority-laden symbols toward affective, entertainment-oriented audiovisual elements reflects creators' identity negotiation to meet algorithmic incentives and user expectations. Yet the information-strategy layer exhibits a subtler balance: more than half of the videos (55.3%) retain Content Rational Appeal Strength, and Content Information Complexity remains moderately high ($M=0.578$). These patterns suggest that, while embracing platform culture, grassroots officials do not fully relinquish their informational role as public servants. For consistency with the count-model specification, Table 3 reports likes, comments, and shares as raw platform-visible counts. Fig. 5 separately presents log-transformed engagement counts only for visual readability under extreme right skew; these transformed values are not used as the negative binomial regression outcomes.

The correlation matrix visualized in Fig. 6 reports pairwise correlations among the 17 modeled independent variables summarized in Table 3. These variables include the nine ELM-anchored strategy constructs listed in Table 1, together with two additional audio/presentation features (Background Music and Speech Rate), one expression-quality indicator, and the video-, account-, publisher-, and region-level controls used in the regression models. The matrix therefore supports the subsequent modeling strategy by showing generally weak-to-moderate correlations among the modeled variables, thereby alleviating multicollinearity concerns. It also clarifies theoretically meaningful relationships among strategy constructs, modeled features, and covariates; for example, Content Rational Appeal Strength correlates positively with Video Duration ($r=0.410$) and Speech Rate ($r=0.400$), suggesting that more rational presentations tend to appear in longer and more verbally dense videos. A notable pattern is the "decoupling" of Source Official Identity Salience from most other modeled variables, with correlations close to zero overall, suggesting that personified authority functions as a relatively independent visual cue.

5.1. Main Effects Analysis

This section examines which strategy combinations are most strongly associated with user engagement across different levels of cognitive effort. To examine these strategy–engagement relationships, we estimate a set of nested regression models grounded in ELM's dual-route framework. The dependent variables (likes, comments, shares) are non-negative count outcomes with pronounced right skew and over-dispersion, making a count-data specification more appropriate than linear regression. The NB models therefore

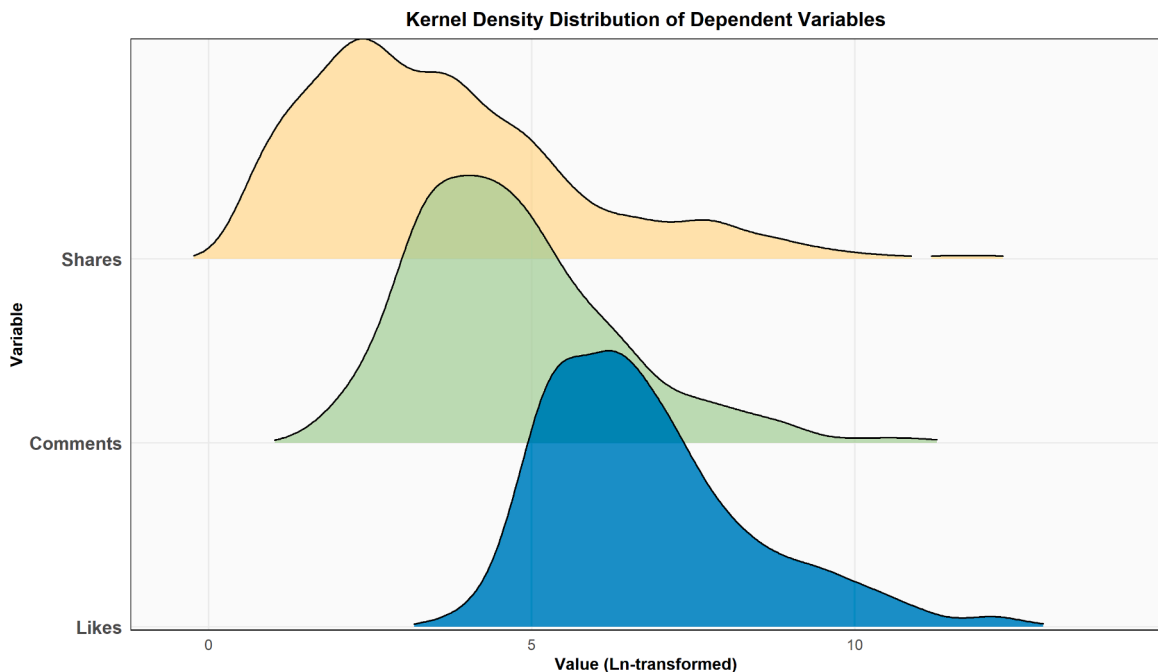


Fig. 5. Kernel Density Distribution of Log-Transformed Engagement Counts. Note: The horizontal axis shows log-transformed engagement counts for visual readability under right skew; regression models use raw non-negative counts as dependent variables.

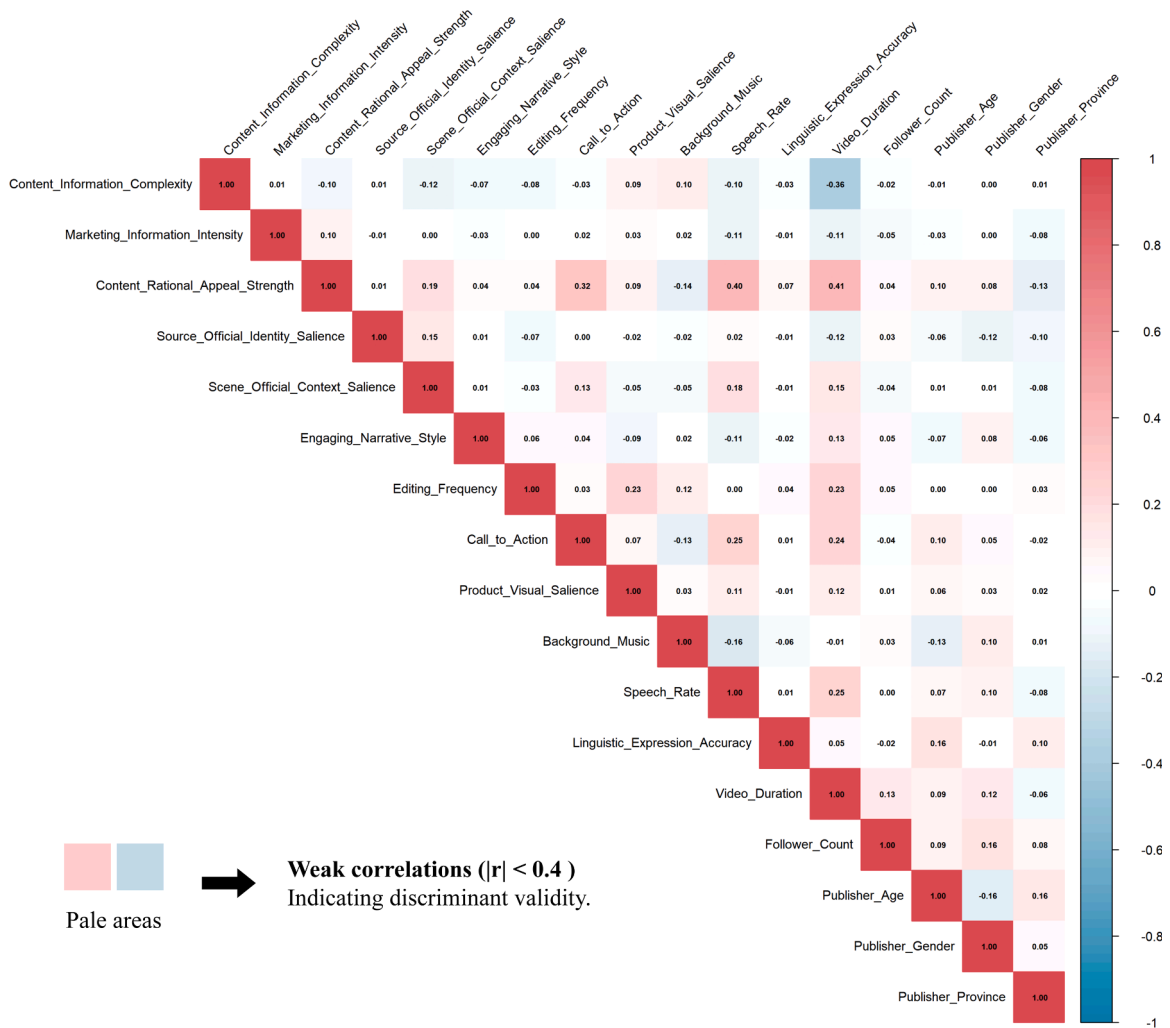


Fig. 6. Correlation Matrix Heatmap of Modeled Independent Variables. **Note:** The matrix includes the 17 independent variables used in the regression models, covering the nine ELM-anchored strategy constructs and the additional features and controls listed in Table 3.

use the raw non-negative counts of likes, comments, and shares, not the log-transformed values shown descriptively in Fig. 5. Specifically, each model estimates expected engagement counts under a log link, with over-dispersion handled through the NB variance structure. Consequently, conventional OLS assumptions are violated, rendering standard linear estimates potentially biased (Ver Hoef & Boveng, 2007). We therefore adopt Negative Binomial (NB) regression as our primary specification. We also attempted Zero-Inflated Negative Binomial (ZINB) models, but they did not converge across multiple reasonable specifications, suggesting that a distinct zero-inflation process is not supported by the data; accordingly, we retain the standard NB model to address over-dispersion. In the main models, we report standard errors clustered at the publisher level. As robustness checks, Appendix Table A4 additionally reports NB estimates with HC3-robust (sandwich) standard errors and PPML results, which remain consistent with the main findings (MacKinnon & White, 1985).

To mitigate algorithmic confounds, we implement a multi-dimensional control strategy. First and foremost, we explicitly control for Follower Count. In the context of algorithmic distribution, this variable serves as a structural proxy for the account's baseline "cold start" traffic pool (Chen & Shi, 2025). By holding this variable constant, we estimate strategy-engagement associations net of pre-existing social capital that supports initial visibility.

Furthermore, we introduce Video Duration as a critical covariate. Given that recommendation systems prioritize metrics such as completion rate—which are intrinsically correlated with length—this control helps distinguish multimodal-cue associations from the mechanical biases associated with video format. Finally, by including Publisher Province and demographic characteristics, we account for observed sources of heterogeneity arising from regional digital disparities. This specification reduces the risk that observed engagement differences reflect structural, temporal, or environmental advantages.

The regression results (Table 4) clarify observed strategy-engagement patterns in grassroots officials' persuasion practices on digital platforms, yielding four principal findings. In interpreting these results, we retain the route classifications established before

Table 4

Negative Binomial Regression Results for Communication Strategies and Public Engagement (N=779).

Variable Classification	Variables	Likes (M1)	Likes (M2)	Comments (M3)	Comments (M4)	Shares (M5)	Shares (M6)
Central Route (ELM)	Content Information	0.160** (0.065)	0.156** (0.063)	0.133* (0.068)	0.151** (0.066)	0.078* (0.042)	0.067** (0.032)
	Complexity	0.019 (0.053)	-0.084* (0.051)	0.165** (0.065)	-0.004 (0.051)	0.158*** (0.026)	0.138*** (0.039)
	Marketing Information	0.019 (0.053)	-0.084* (0.051)	0.165** (0.065)	-0.004 (0.051)	0.158*** (0.026)	0.138*** (0.039)
	Intensity	0.019 (0.053)	-0.084* (0.051)	0.165** (0.065)	-0.004 (0.051)	0.158*** (0.026)	0.138*** (0.039)
Peripheral Route (ELM)	Content Rational Appeal	-0.563*** (0.058)	-0.307*** (0.060)	-0.671*** (0.060)	-0.327*** (0.061)	-0.883*** (0.040)	-0.515*** (0.044)
	Strength	-0.563*** (0.058)	-0.307*** (0.060)	-0.671*** (0.060)	-0.327*** (0.061)	-0.883*** (0.040)	-0.515*** (0.044)
	Source Official Identity	-0.563*** (0.058)	-0.307*** (0.060)	-0.671*** (0.060)	-0.327*** (0.061)	-0.883*** (0.040)	-0.515*** (0.044)
	Saliency	-0.563*** (0.058)	-0.307*** (0.060)	-0.671*** (0.060)	-0.327*** (0.061)	-0.883*** (0.040)	-0.515*** (0.044)
	Scene Official Context Saliency	-0.563*** (0.058)	-0.307*** (0.060)	-0.671*** (0.060)	-0.327*** (0.061)	-0.883*** (0.040)	-0.515*** (0.044)
	Engaging Narrative Style	-0.563*** (0.058)	-0.307*** (0.060)	-0.671*** (0.060)	-0.327*** (0.061)	-0.883*** (0.040)	-0.515*** (0.044)
	Editing Frequency	-0.563*** (0.058)	-0.307*** (0.060)	-0.671*** (0.060)	-0.327*** (0.061)	-0.883*** (0.040)	-0.515*** (0.044)
	Call to Action	-0.563*** (0.058)	-0.307*** (0.060)	-0.671*** (0.060)	-0.327*** (0.061)	-0.883*** (0.040)	-0.515*** (0.044)
	Product Visual Saliency	-0.563*** (0.058)	-0.307*** (0.060)	-0.671*** (0.060)	-0.327*** (0.061)	-0.883*** (0.040)	-0.515*** (0.044)
	Background Music	-0.563*** (0.058)	-0.307*** (0.060)	-0.671*** (0.060)	-0.327*** (0.061)	-0.883*** (0.040)	-0.515*** (0.044)
	Speech Rate	-0.563*** (0.058)	-0.307*** (0.060)	-0.671*** (0.060)	-0.327*** (0.061)	-0.883*** (0.040)	-0.515*** (0.044)
	Linguistic Expression	-0.563*** (0.058)	-0.307*** (0.060)	-0.671*** (0.060)	-0.327*** (0.061)	-0.883*** (0.040)	-0.515*** (0.044)
	Accuracy	-0.563*** (0.058)	-0.307*** (0.060)	-0.671*** (0.060)	-0.327*** (0.061)	-0.883*** (0.040)	-0.515*** (0.044)
	Video Duration	-0.563*** (0.058)	-0.307*** (0.060)	-0.671*** (0.060)	-0.327*** (0.061)	-0.883*** (0.040)	-0.515*** (0.044)
	Follower Count	-0.563*** (0.058)	-0.307*** (0.060)	-0.671*** (0.060)	-0.327*** (0.061)	-0.883*** (0.040)	-0.515*** (0.044)
	Publisher Age	-0.563*** (0.058)	-0.307*** (0.060)	-0.671*** (0.060)	-0.327*** (0.061)	-0.883*** (0.040)	-0.515*** (0.044)
Control Variables	Publisher Gender	-0.220*** (0.054)	-0.101* (0.052)	-0.345*** (0.056)	-0.125** (0.052)	-0.478*** (0.044)	-0.353*** (0.054)
	Publisher Province	-0.035 (0.053)	0.004 (0.051)	-0.118** (0.056)	-0.107** (0.054)	-0.212*** (0.041)	-0.188*** (0.046)
	Intercept	8.083*** (0.051)	7.914*** (0.048)	6.097*** (0.054)	5.836*** (0.049)	5.971*** (0.043)	4.842*** (0.054)
	Pseudo-R ²	0.301	0.391	0.275	0.406	0.235	0.409

Note: Dependent variables are raw counts of likes, comments, and shares. Standard errors are in parentheses. *p < 0.1; **p < 0.05; ***p < 0.01.

modeling in Table 2; coefficient signs are used to assess whether a theoretically assigned cue is positively or negatively associated with engagement, not to determine its ELM route.

First, the central-route results indicate a striking paradox in which substantive information outperforms traditional didacticism. Content Information Complexity is consistently and positively associated with all forms of engagement, suggesting that audiences differentiate substantive cognitive value from bureaucratic didacticism and are willing to invest cognitive resources to endorse high-quality information. Although this variable is positive across all engagement metrics, its estimated coefficient remains numerically lower for shares ($\beta=0.067$, $p<0.05$) than for likes ($\beta=0.156$, $p<0.05$) and comments ($\beta=0.151$, $p<0.05$).

Notably, Marketing Information Intensity exhibits a robust positive association specifically with sharing ($\beta=0.138$, $p<0.01$) while showing no significant relationship with likes or comments, contrasting sharply with the consistent negative associations of Content Rational Appeal Strength. This nuance is critical: audiences do not appear to reject detailed product information (Intensity) or high-value knowledge (Complexity); rather, the pattern suggests lower engagement for the didactic, prescriptive tone associated with "Rational Appeal." Thus, successful central-route persuasion appears to benefit from decoupling high-value information from formulaic bureaucratic framing.

Second, authority, a canonical peripheral cue, exhibits a pronounced bifurcation in observed engagement patterns, which is contingent on personalization rather than institutionalization. This finding, summarized as "authority resides in persons, not places," is evidenced by divergent behavioral outcomes. Authority personified through an official's attire (Source Official Identity Saliency) acts as a powerful trust proxy that is positively associated with engagement. In contrast, its institutionalized expression through bureaucratic backdrops (Scene Official Context Saliency) is negatively associated with interaction and is consistent with psychological-distance interpretations. This supports the interpretation that effective identity negotiation retains the personalized carrier of authority while stripping away its colder institutional context.

Third, the utility of peripheral cues follows a differentiated spectrum centered on experience enhancement versus intention revelation. Not all peripheral tactics are beneficial. Audiences distinguish experience-enhancing cues (e.g., Engaging Narrative Style, high Editing Frequency) from intention-revealing cues (e.g., explicit Call to Action, overly strong Product Visual Saliency). The former

carry hedonic value that is associated with higher engagement by optimizing sensory and affective experience (Hughes et al., 2019; Pan et al., 2025). By contrast, intention-revealing cues—such as Call to Action ($\beta=-0.286$, $p<0.01$) and Product Visual Salience ($\beta=-0.318$, $p<0.01$)—make commercial or persuasive intent overly explicit, a pattern consistent with defensive responses and lower engagement (Agarwal et al., 2025).

Finally, a stepwise logic underlies the observed engagement patterns at different depths. From the baseline to the extended models, adding audiovisual peripheral strategies substantially increases explanatory power for likes, comments, and shares (Pseudo- R^2 increases by 29.9%, 47.6%, and 74.0% respectively), indicating that peripheral cues account for a large share of variance in user engagement. More importantly, the observed associations diverge by engagement depth. Low-cost interactions such as likes are more strongly associated with a set of experience-oriented peripheral cues, whereas sharing, the highest-cost form of deep endorsement, is associated with a more demanding combination: strong source credibility (high Source Official Identity Salience), deep affective resonance (high Engaging Narrative Style), and exceptional informational value (high Content Information Complexity) acting in concert.

5.2. Robustness Checks and Heterogeneity Analysis

To examine the boundary conditions of our findings, we assessed how video duration, account size, and regional context condition the association between persuasion strategies and sharing behavior. We further estimated full-sample negative binomial interaction models by interacting each subgroup indicator with the strategy variables; the interaction-based coefficient-difference tests are reported in Appendix Table A7 and provide statistical evidence for several key heterogeneity patterns emphasized below. To assess the statistical robustness of our findings, we implemented a multidimensional validation strategy (see Appendix Tables A3–A5 and A7).

First, regarding model specification, we tested Zero-Inflated Negative Binomial (ZINB) models (Hilbe, 2011); however, their persistent non-convergence did not provide stable evidence for a distinct zero-inflation process in this dataset, so we retained the standard Negative Binomial specification. Second, to address potential heteroskedasticity and estimator dependence, we re-estimated the models using the Poisson Pseudo-Maximum Likelihood (PPML) estimator (Silva & Tenreiro, 2006). As shown in Table A4, the results remained broadly consistent with the main findings in sign and substantive interpretation.

Third, to mitigate observable selection bias and assess robustness to observed-covariate imbalance, we used Propensity Score Matching (PSM) as a robustness check to estimate the ATT with logit-based propensity scores and nearest-neighbor matching (Rosenbaum & Rubin, 1983). For each strategy treatment, propensity scores used the same observable video-, publisher-, account-, regional-, and expression-quality controls as the main models: video duration, publisher gender, publisher age, follower count, publisher province, and linguistic expression accuracy. These covariates capture observable differences in content form, publisher characteristics, account scale, regional context, and expression quality that could jointly shape strategy adoption and engagement. Table A5 supports the robustness of the estimated associations after adjustment for observed-covariate imbalance, and Appendix Fig. A3 reports treatment-specific SMD diagnostics showing that post-matching imbalance is substantially reduced and falls within or close to the conventional $\pm 10\%$ threshold. Table A6 summarizes the analytical transparency and alternative validation logic.

Table 5
Heterogeneity Analysis of Shares by Video Duration, Account Size, and Regional Governance Context (Negative Binomial Regression).

Variable Classification	Variables	Video Long	Video Short	Fans Large	Fans Small	Gov Good	Gov Poor
Central Route	Content Information Complexity	0.227*** (0.064)	-0.145* (0.082)	-0.176* (0.090)	0.458*** (0.102)	-0.217*** (0.061)	0.772*** (0.135)
	Marketing Information Intensity	-0.260*** (0.083)	0.140* (0.075)	0.035 (0.070)	-0.134** (0.065)	0.145* (0.076)	-0.246*** (0.074)
	Content Rational Appeal Strength	-0.389*** (0.044)	-0.212*** (0.052)	-0.333*** (0.082)	-0.214*** (0.043)	-0.205*** (0.073)	-0.865*** (0.087)
Peripheral Route	Source Official Identity Salience	0.847*** (0.155)	0.128*** (0.048)	-0.082* (0.045)	0.746*** (0.183)	-0.082* (0.046)	0.768*** (0.135)
	Scene Official Context Salience	-0.209*** (0.045)	-0.270*** (0.074)	-0.172** (0.075)	-0.425*** (0.044)	-0.215*** (0.075)	0.128*** (0.042)
	Engaging Narrative Style	-0.127*** (0.034)	0.066 (0.052)	0.168* (0.090)	0.286*** (0.039)	0.195*** (0.070)	0.107*** (0.029)
	Editing Frequency	0.507*** (0.036)	0.200*** (0.061)	0.196** (0.083)	0.423*** (0.046)	0.148** (0.075)	0.525*** (0.077)
	Call to Action	-0.446*** (0.064)	-0.148* (0.078)	-0.107 (0.077)	-0.209*** (0.033)	-0.195*** (0.070)	-0.175*** (0.058)
	Product Visual Salience	-0.560*** (0.034)	-0.353*** (0.072)	-0.210*** (0.075)	-0.402*** (0.043)	-0.210*** (0.057)	-0.863*** (0.099)
	Background Music	0.357*** (0.031)	0.200*** (0.070)	0.350*** (0.066)	0.198*** (0.042)	0.240*** (0.060)	0.208*** (0.034)
	Speech Rate	-0.284*** (0.063)	-0.145*** (0.042)	-0.578*** (0.077)	-0.186*** (0.061)	-0.413*** (0.083)	-0.283*** (0.071)
	Controls	Included	Yes	Yes	Yes	Yes	Yes
Model Fit	Pseudo- R^2	0.558	0.323	0.426	0.532	0.393	0.401

Note: Standard errors are in parentheses. Subgroups use median splits for Video Duration/Follower Count and categorical Gov Good/Poor; Table A7 reports interaction tests. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

To identify theoretical boundary conditions, we stratified the heterogeneity analysis by video duration, account follower size, and regional digital governance level, using shares as the dependent variable because sharing represents the highest threshold of user endorsement and social reputation commitment (Table 5).

First, Video Duration shows a theoretical divergence consistent with temporal constraints on central-route processing. The two central-route strategies exhibit inverse patterns: Marketing Information Intensity is positively associated with sharing in short videos ($\beta=0.140$, $p<0.1$) but negatively associated with sharing in longer formats ($\beta=-0.260$, $p<0.01$), suggesting that prolonged exposure to dense sales cues may trigger persuasion resistance. Conversely, Content Information Complexity is positively associated with sharing in longer videos ($\beta=0.227$, $p<0.01$) but marginally negative in short videos ($\beta=-0.145$, $p<0.1$). This pattern is consistent with the idea that central-route persuasion is more plausible when sufficient cognitive opportunity exists: in videos under 30 seconds, efficient product cues are easier to process, whereas higher complexity may become a cognitive burden.

Second, the analysis of Follower Count is consistent with a "legitimacy compensation" pattern. For smaller accounts ("Fans Small"), strong authority cues—specifically Source Official Identity Salience ($\beta=0.746$, $p<0.01$) and Content Information Complexity ($\beta=0.458$, $p<0.01$)—are strongly associated with sharing. Conversely, for established influencers ("Fans Large"), these cues are largely ineffective or even counterproductive (Identity: $\beta=-0.082$, $p<0.1$; Complexity: $\beta=-0.176$, $p<0.1$). This suggests that grassroots officials with lower social capital appear to rely more on "hard" institutional symbols to compensate for a lack of initial credibility, whereas established accounts appear more vulnerable to an "authenticity penalty" where such formal cues distance them from their audience.

Third, the Regional Governance Context is consistent with an informational scarcity pattern. In regions with lagging digital governance development, authoritative and complex information is more strongly associated with sharing (Complexity: $\beta=0.772$, $p<0.01$; Identity: $\beta=0.768$, $p<0.01$). Crucially, even institutionalized settings—which are typically negatively associated with engagement—show a positive association in this context (Scene Official Context: $\beta=0.128$, $p<0.01$). This pattern suggests that in information-scarce environments, audiences prioritize the credibility signaled by official backdrops over platform-native aesthetics, treating official videos as scarce sources of reliable public data. In contrast, in regions with advanced digital governance, where users are likely saturated with official messaging, these traditional signals are negatively associated with sharing (Complexity: $\beta=-0.217$, $p<0.01$; Identity: $\beta=-0.082$, $p<0.1$), indicating a higher demand for novel, platform-native presentation styles over redundant officialness (Xu et al., 2025).

6. Discussion

This study suggests that, on algorithmically driven short-video platforms, observable engagement with government communication does not hinge on optimizing a single persuasive route. Rather, the findings are consistent with a pattern in which multidimensional strategy cues drawn from both the central and peripheral routes align with the cost ladder of public engagement. The following sections detail the empirical findings, their theoretical contributions to information management and social science, and actionable strategies for practitioners.

6.1. Summary of Key Findings

First, the central-route pattern shifts from traditional didactic rationality to a coupling of cognitive value and narrative experience. High informational density, especially Marketing Information Intensity, is positively associated with engagement mainly in efficient short-form contexts, whereas stereotypically didactic "Rational Appeal" is consistently negatively associated with engagement, especially for sharing. In an attention-scarce environment, audiences do not reject valuable information; they appear to reject monotonous, high-load presentation. High-value information therefore appears more likely to remain engaging when embedded in narrative and sensory experiences (Li et al., 2026). Experience-enhancing peripheral cues, especially Engaging Narrative Style and Background Music, may provide a cognitive escort by increasing affective immersion, reducing immediate counter-arguing (Li et al., 2024), and increasing perceived social presence and parasocial familiarity with the official communicator (H.-J. Kim et al., 2025).

Second, the analysis supports a "bifurcation of authority," where the engagement value of officialdom appears to hinge on its mode of presentation rather than its mere presence. While our results indicate that personified cues (e.g., uniforms) act as effective trust proxies, institutionalized contexts (e.g., bureaucratic offices) are negatively associated with engagement. This divergence can be explained through the lens of social identity theory: personified authority cues facilitate the recategorization of the official as a proximate, approachable community member, whereas static institutional backdrops accentuate bureaucratic category boundaries and amplify "outgroup" perception (Peña et al., 2026). Effective identity negotiation, therefore, depends on retaining the individualized carrier of authority while stripping away cues that foreground institutional distance.

Third, the utility of peripheral cues follows a differentiated spectrum centered on experience enhancement versus intention revelation. Not all peripheral tactics are beneficial. Audiences distinguish experience-enhancing cues (e.g., Engaging Narrative Style, high Editing Frequency) from intention-revealing cues (e.g., explicit Call to Action, overly strong Product Visual Salience). The former carry hedonic value that is associated with higher engagement by optimizing sensory and affective experience (Hughes et al., 2019; Pan et al., 2025); the latter prematurely and excessively expose persuasive or commercial intent, readily activating persuasion knowledge, vigilance and defense against perceived attempts to influence, and thus backfiring (Agarwal et al., 2025).

Fourth, these patterns cohere in a unified dynamic-matching framework in which strategy–engagement associations depend on their alignment with the costs of engagement behaviors (the "Cost Ladder"). Likes, which represent low-effort feedback (Yang et al., 2020), are more strongly associated with experience-enhancing peripheral cues. Comments require greater emotional involvement and cognitive effort: users seek to discuss content with informational depth but resist one-way didactic messaging (Garz et al., 2020).

Shares, the highest-cost form of deep endorsement, consume cognitive resources and function as social backing that puts personal reputation at stake. Crossing this stricter persuasive threshold is associated with a concerted bundle: personalized authority (high Source Official Identity Salience), strong affective resonance (high Engaging Narrative Style), and exceptional cognitive value (high Content Information Complexity) acting in synergy consistent with transitions from passive reception to active diffusion (Deng et al., 2025; Li et al., 2024).

Consistent with the interaction-based coefficient-difference checks reported in Appendix Table A7, our heterogeneity analysis (Table 5) provides evidence for several association-based boundary patterns that characterize this synergistic matching. First, temporal constraints shape the feasibility of central-route processing; higher Information Complexity is positively associated with sharing in longer videos (>30s) where users have sufficient opportunity for elaboration, but corresponds to lower sharing in shorter clips due to cognitive overload (Nguyen et al., 2025). Second, the follower-count results are consistent with a legitimacy compensation pattern. Smaller accounts rely more heavily on formal symbols such as uniforms to establish initial, institution-based trust, whereas larger and more established accounts incur an authenticity penalty in which the same symbols reduce engagement (Alibakhshi & Srivastava, 2025). This pattern aligns with stage-contingent trust formation (Y. A. Kim & Ahmad, 2013): early-stage accounts lacking interaction history appear to rely more on "hard" institutional anchors to secure initial credibility, whereas mature accounts thrive on experience-based relational trust, making rigid formality incompatible with the accumulated expectations of authenticity. Third, information scarcity modulates value perception; in regions with lower digital governance development, authoritative and complex content is more strongly associated with sharing (Jiang et al., 2025), a pattern that appears weaker or reversed in saturated, high-governance environments.

6.2. Theoretical Implications

This study bridges the epistemological gap between computational efficiency and theoretical depth, offering distinct contributions to research perspectives, methodological protocols in Information Management, and digital persuasion theory in Social Science.

6.2.1. Shifting Research Perspectives: From Static Text to Dynamic Multimodality

Traditional government information research has predominantly relied on text mining of policy documents or social media posts (Qin et al., 2025). This study supports a shift in research perspective toward dynamic multimodal analysis. The persuasive force of short video may not arise from a static bundle of multimodal cues alone, but from their temporal evolution and cumulative patterns (Sim et al., 2024). By operationalizing constructs such as "dynamic contrastive" visual changes (e.g., the shift from office to field), we show that observed persuasion-related patterns can be interpreted as temporally patterned, defined not just by static visual elements, but by the temporal sequence of sensory stimuli. This challenges the "bag-of-features" assumption prevalent in previous computer-mediated communication research and supports the importance of temporal dynamics for interpreting platformized digital governance communication.

6.2.2. Methodological Contributions: A Human-AI Symbiosis Protocol

For the field of Information Management, the AI-enhanced analytical framework proposed and validated here establishes a rigorous and operable paradigm for computational social science in the era of generative AI. As visualized in Fig. 2, this protocol addresses the "black box" problem of MM-LLM through core methodological principles:

First, it operationalizes "cognitive complementarity" to redefine reliability standards. In the AI era, reliability must extend beyond human-human agreement to encompass a plural verification system. As evidenced by the reliability diagnostics in Fig. 4, AI demonstrates superior consistency on patterned visual features, whereas humans excel when contextual understanding and affective inference are required. Consequently, our workflow allocates structured pattern recognition to AI and conceptual adjudication to humans, maximizing both scalability and validity.

Second, it enforces structural "auditability" for unstructured data. Given the inherent opacity of foundation models, traditional code-and-data replication is insufficient to guarantee credibility. Our framework necessitates a clear, traceable audit trail from raw data to final claims, mandating systematic documentation of decision rationales at every step of human-AI interaction. This creates a standard for procedural transparency essential for the rigorous replication of future video-based studies (Naeem et al., 2025).

6.2.3. Advancing Digital Persuasion Theory

In Social Science, this study offers a bounded extension of the Elaboration Likelihood Model (ELM) for algorithmic governance. Rather than proposing a separate persuasion model, it specifies how ELM's central and peripheral routes can be interpreted when cues are multimodal, temporally sequenced, and embedded in platformized relationships. The central insight is that authority as a peripheral cue is bifurcated: its association with engagement depends less on "authority" per se than on whether authority is personified or institutionalized. The "Cost Ladder" refinement further suggests that central and peripheral cues are complementary rather than competitive: experience-enhancing peripheral cues can be interpreted as the "entry ticket" for low-cost engagement, while high-value central cues serve as the "currency" for high-cost endorsement. This refinement shifts ELM application from static cue classification to temporally informed cue interpretation, from isolated cues to cue configurations, and from purely cognitive processing to processing conditioned by social identity, perceived social presence, and trust formation. Thus, ELM remains a useful organizing framework in algorithmic governance, but its explanatory value improves when paired with identity-based boundary processes and stage-contingent trust formation.

6.3. Practical Implications

For local-government practitioners and content creators who operate in comparable platformized governance-marketing contexts and navigate the "official-influencer" dilemma, this study offers evidence-bounded guidance on identity construction, strategy matching, and lifecycle management. First, creators can address the central tension of their hybrid role by strategically "decoupling authority from bureaucracy." The empirical evidence suggests that while authority remains a potent trust proxy, its observed association with engagement varies by mode of presentation. Officials may emphasize personified authority cues, such as wearing official attire or insignia, which can establish immediate credibility without alienating the audience. Simultaneously, they may avoid over-reliance on institutionalized contexts, such as filming purely within static government offices, which are negatively associated with engagement in our models and may create psychological distance. By positioning themselves as visible representatives who are nonetheless accessible individuals, practitioners may better align professional legitimacy with the authenticity and intimacy norms of comparable algorithmically mediated short-video environments.

Furthermore, in comparable algorithmically mediated short-video environments, content strategies can be calibrated to the "cost ladder" of public engagement to support communicative effectiveness. Officials may clarify their engagement goals and deploy distinct cue combinations accordingly. To encourage broad, low-cost visibility such as likes, creators can emphasize experience-enhancing cues, leveraging rhythmic editing, background music, and narrative twists to support sensory immersion. Conversely, to elicit deep social diffusion in the form of shares, content appears to benefit from high Information Complexity and substantive cognitive value rather than empty rhetoric. Crucially, this information delivery is more likely to remain engaging when decoupled from the didactic tone often associated with "rational preaching," which is negatively associated with engagement in our models (see Table 4, M4). Instead, high-value information can be embedded within conversational narratives to reduce defensive processing, so that the audience perceives the content as valuable knowledge rather than administrative instruction.

Finally, the deployment of authority cues can be adapted according to the account's lifecycle stage. Newer or smaller official accounts often face a "legitimacy deficit" and may place greater emphasis on "hard" symbols, such as uniforms and badges, to draw on the observed "legitimacy compensation" pattern for initial trust building (cold start). However, as the account matures and accumulates social capital, these formal symbols can be gradually relaxed. Continued over-reliance on rigid authority markers in established accounts may be associated with an "authenticity penalty," where excessive formality distances the creator from an audience that has already granted them trust based on personal connection.

Therefore, while established accounts should transition toward a personal digital brand, for resource-limited practitioners who may find the full research workflow challenging, we recommend a **Simplified Application Workflow**. Rather than strictly replicating the inductive theory-building phases, government media teams can implement a three-step protocol: (1) *Adaptation*: modify the validated codebook (Table 2) to local policy contexts via prompt engineering; (2) *Calibration*: refine prompts on a small pilot sample to verify output consistency; and (3) *Deployment*: execute a "Direct-Inference Mode" using cost-effective models (e.g., Gemini Flash). This approach renders high-precision monitoring accessible to grassroots departments with minimal computational overhead.

7. Conclusions, Limitations, and Future Directions

7.1. Conclusions and Contributions

This study makes a dual contribution to Information Management and Government Communication by examining Chinese grassroots officials' D2C marketing videos on Douyin. Theoretically, it systematically examines persuasion patterns of "government official influencers" within this platformized short-video setting by identifying the "bifurcation of authority" and the "legitimacy compensation" pattern. Crucially, this study refines the application of the Elaboration Likelihood Model to short-video environments by specifying how central and peripheral cues can be interpreted under temporal and multimodal conditions. We find evidence that, within this algorithmically mediated setting, persuasion can be interpreted as a dynamic matching process rather than simply a fixed accumulation of cues. Specifically, the observed associations of authority and affect are not intrinsic to the cue itself but vary with the temporal "cost ladder" of engagement, with deeper platform-visible engagement associated with higher-order cue combinations.

Methodologically, we develop and implement an AI-enhanced analytical framework that addresses the trade-off between theoretical depth and scalability in large-scale multimodal video analysis. In contrast to studies that use models primarily as automated classifiers for feature counting, our protocol specifies a structured, theory-informed human-in-the-loop procedure for extracting and validating multimodal cues. This design supports the interpretation of complex semantic relationships rather than simple signal detection, enabling rigorous analysis under the constraints of rapidly evolving platform media.

7.2. Limitations and Future Directions

Notwithstanding these contributions, the conclusions should be read within specific boundary conditions that, in turn, delineate promising avenues for further inquiry. First, while the framework relies on generative AI, our cross-model validation (Appendix Fig. A2) indicates that the identified strategies are robust to variations in model parameter size. This consistency suggests that the framework possesses "instrumental resilience," where reliability derives from the structured "Context-Injection" protocol and rigorous construct definitions rather than the idiosyncratic weights of a specific foundation model. While this diagnostic check supports the protocol's robustness, future work could further extend this validation across heterogeneous model families (e.g., Qwen or Llama) to isolate potential algorithmic biases inherent to specific architectures. Furthermore, we anticipate that emerging models

trained on broader multimodal corpora will likely yield enhanced consistency and a more refined capacity to decode the subtle, human-like attentional cues that underpin persuasion.

Second, the empirical setting, China’s grassroots governance on Douyin, limits generalizability, especially regarding cultural contingencies and platform-induced biases. Although the tension between bureaucratic and platform logics may recur across contexts, cultural dimensions such as power distance can shape how audiences interpret institutional symbols and authority cues (Daniels & Greguras, 2014). Douyin should therefore be treated not as a neutral distribution channel but as a platform environment in which recommendation mechanisms, trending dynamics, creator norms, and short-video culture shape which official videos become visible, interacted with, and observable. Because the capped extraction strategy prioritized top-engagement videos within each account, the corpus is well suited to analyzing visible D2C communication but may underrepresent low-exposure, less algorithmically amplified, or less platform-native content. Cross-cultural replication across platforms and political systems is needed to assess when institutional contexts function as legitimacy signals and when they increase psychological distance. Although our models account for observed covariates and overdispersed engagement counts, unobserved real-time algorithmic boosting may still introduce residual confounding. Likewise, likes, comments, and shares should be interpreted as platform-visible behavioral traces rather than direct measures of public attitudes, persuasion outcomes, or offline action; future work could address this limitation through controlled field experiments.

Third, our theoretical framing is organized around ELM, and the quantitative models do not directly operationalize identity-based or relational psychological states. Accordingly, social identity, perceived social presence, and parasocial perspectives are used as interpretive lenses to clarify the observed patterns rather than as empirically measured mechanisms. Future research should directly measure these constructs to test the proposed psychological pathways and to evaluate the strength of these mechanisms across contexts.

Building on these conclusions and boundaries, future research can proceed along three interrelated avenues. On the theoretical front, the observed bifurcation of authority calls for moving beyond traditional source-credibility models toward a dynamic digital source-credibility theory that integrates constructs such as personified authority and social presence. In application, the framework should be deployed for cross-cultural comparative validation. Future research could apply this protocol to datasets from distinct political contexts (e.g., Western officials on TikTok) to empirically verify whether the “Cost Ladder” and “Authority Bifurcation” persist across divergent civic engagement cultures. Methodologically, the most forward-looking direction is to evolve AI from an efficient “measurement tool” into a genuine “partner in theory discovery.” This entails systematically designing experiments and analytical workflows to explore and verify the potential of large multimodal models for abductive reasoning and original theory building, an agenda that would both deepen understanding of complex social phenomena and help reshape the research paradigm of computational social science.

Ethical approval

Our study did not require further ethics committee approval as it did not involve animal or human clinical trials and was not unethical.

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CRedit authorship contribution statement

Yuying Luo: Writing – original draft, Visualization, Validation, Resources, Investigation, Formal analysis, Data curation, Conceptualization. **Xue Lei:** Validation, Supervision, Resources, Project administration, Investigation. **Wenxuan Zhang:** Validation, Software, Methodology, Conceptualization. **Hao Zheng:** Validation, Supervision, Investigation, Data curation. **Xueguo Xu:** Validation, Supervision, Funding acquisition, Conceptualization. **Jun Xu:** Validation, Software, Methodology, Data curation, Conceptualization.

Declaration of conflicting interest

The authors declare no competing financial interest.

Appendix

Table A1
Structured Prompting Protocol for the AI-Enhanced Analytical Framework.

Phase	Core Instruction / Prompt Logic	Key Input Variables	Mechanism & Function
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Table A1 (continued)

Phase	Core Instruction / Prompt Logic	Key Input Variables	Mechanism & Function
1. Exploratory Feature Emergence	Role: Cognitive Visual Analysis Expert (guided by LC4MP). Task: Perform frame-by-frame analysis covering: (1) Pre-Attentive Processing, (2) Attention Allocation, (3) Information Encoding. Constraint: Focus on dynamic changes relative to $t - 1$ rather than static description.	author_profile video_description key_frame_image	Simulates open-coding to generate atomic, descriptive features for inductive analysis.
2. Inductive Strategy Interpretation	Task: Evaluate topic cluster validity (Coherence/Relevance on 1-5 scale). Logic: Classify based on ELM and "Identity Orientation" (Authority vs. Engagement). Output: Recommendation (Accept/Review/Reject) with academic justification.	topic_keywords representative_docs coherence_score	Bridges unsupervised clustering (BERTopic) with theoretical filtering to remove noise.
3. Theoretical Construct Synthesis	Protocol: Three-Step Deduction. Step 1: Empirical Deconstruction (Actors, Context, Tone). Step 2: Thematic Abstraction (Hypothesize persuasion mechanism). Step 3: Operationalization (Map to ELM route; define binary/continuous code).	topic_summary validation_scores	Abductive reasoning to transform raw patterns into measurable, theoretically grounded constructs.
4. Large-Scale Quantitative Coding	Role: Quantitative Content Analyst. Protocol: Sequential Context-Injection. 1. Ingest Audio/Meta -> 2. Code Visual Frames -> 3. Holistic Synthesis. Constraint: Strict adherence to codebook; "Consistency Mandate" applies.	video_metadata audio_transcript key_frame_sequence	Automated application of the finalized codebook with maximum reproducibility (Temp=0.0).

Table A2

Technical Specifications and Prompt Engineering Mechanics.

Component	Configuration / Core Template Structure
1. Model Hyperparameters	Model: Gemini 2.5 Pro Temperature: 0.0 (Measurement Phase); 0.4 (Discovery Phase) Top-p: 0.95 Safety Settings: BLOCK_ONLY_HIGH
2. System Persona	Instruction: "You are a quantitative content analysis assistant. Apply the codebook exactly and base all decisions only on the provided evidence."
3. Audio-First Context Injection	Logic: Establish semantic baseline before visual decoding. Template: "Input: [Audio_Transcript] + [Metadata]. Task: Analyze this context to infer the core product category and emotional tone. Await response before processing visual frames."
4. Dynamic Contrastive Protocol	Logic: Differential Analysis (t vs. $t - 1$). Template: "Frame Timestamp: [T]. Context: [Previous_Frame_Analysis]. Instruction: Identify strictly what is NEW or DIFFERENT in visual elements/narrative compared to the previous frame."
5. Chain-of-Thought (CoT)	Logic: Internal Reasoning Trace required for all variables. Template: "For variable [X], perform silent reasoning: (1) Check definition, (2) Scan evidence, (3) Determine score. Output: JSON format only."

Table A3

Robustness Check using OLS (N=779).

Variable Classification	Variables	Likes (M1)	Likes (M2)	Comments (M3)	Comments (M4)	Shares (M5)	Shares (M6)
Central Route (ELM)	Content Information	0.757*** (0.283)	0.751*** (0.272)	0.728** (0.350)	0.649* (0.377)	0.945** (0.433)	0.853** (0.391)
	Marketing Information	-0.105 (0.339)	-0.367 (0.328)	0.548 (0.348)	0.206 (0.332)	1.350*** (0.484)	0.975*** (0.371)
	Intensity	-0.803*** (0.119)	-0.364*** (0.124)	-0.892*** (0.122)	-0.385*** (0.126)	-0.971*** (0.170)	-0.424** (0.179)
	Content Rational Appeal						
	Strength						
Peripheral Route (ELM)	Source Official Identity		0.213* (0.116)		0.200* (0.118)		0.348** (0.167)
	Salience						
	Scene Official Context Salience		-0.320** (0.145)		-0.372** (0.147)		-0.401** (0.194)
	Engaging Narrative Style		0.203* (0.121)		0.201* (0.122)		0.375** (0.173)
	Editing Frequency		0.167 (0.128)		0.104 (0.130)		0.354* (0.185)
	Call to Action		-0.476*** (0.110)		-0.455*** (0.111)		-0.485*** (0.158)
	Product Visual Salience		-0.311** (0.129)		-0.296** (0.130)		-0.396** (0.185)
	Background Music		0.404**		0.538***		0.524*

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Table A3 (continued)

Variable Classification	Variables	Likes (M1)	Likes (M2)	Comments (M3)	Comments (M4)	Shares (M5)	Shares (M6)
Control Variables	Speech Rate		(0.200) -0.255*** (0.047)		(0.203) -0.317*** (0.048)		(0.288) -0.366*** (0.068)
	Linguistic Expression	-0.096 (0.194)	-0.168 (0.185)	0.012 (0.200)	-0.06 (0.188)	0.023 (0.277)	-0.065 (0.266)
	Accuracy	-0.052 (0.094)	0.028 (0.095)	0.026 (0.097)	0.113 (0.096)	0.071 (0.135)	0.134 (0.137)
	Video Duration	0.482*** (0.040)	0.475*** (0.038)	0.355*** (0.042)	0.342*** (0.040)	0.205*** (0.035)	0.198*** (0.032)
	Follower Count	-0.017** (0.008)	-0.007 (0.008)	-0.016** (0.008)	-0.006 (0.008)	-0.042*** (0.012)	-0.029*** (0.011)
	Publisher Age	0.025 (0.146)	0.118 (0.142)	-0.122 (0.150)	-0.026 (0.144)	-0.535** (0.208)	-0.208 (0.209)
	Publisher Gender	-0.080 (0.066)	-0.081 (0.064)	-0.116* (0.068)	-0.122* (0.064)	-0.311*** (0.095)	-0.309*** (0.091)
	Publisher Province	7.674*** (0.543)	7.563*** (0.581)	5.334*** (0.557)	5.331*** (0.589)	5.078*** (0.774)	4.597*** (0.851)
	Intercept						
	R ²	0.170	0.26	0.133	0.246	0.124	0.205

Table A4

Other model validation & robustness.

Variable Classification	Variables	NB estimates with HC3-robust (sandwich) SEs			PPML Poisson Pseudo-Maximum Likelihood		
		Likes (M1)	Comments (M2)	Shares (M3)	Likes (M4)	Comments (M5)	Shares (M6)
Central Route (ELM)	Content Information Complexity	0.156** (0.071)	0.151* (0.082)	0.067* (0.038)	0.034*** (0.011)	0.053*** (0.015)	0.080*** (0.025)
	Marketing Information Intensity	-0.084 (0.061)	-0.004 (0.063)	0.138*** (0.045)	-0.004 (0.008)	0.014 (0.012)	0.047** (0.019)
	Content Rational Appeal Strength	-0.307*** (0.080)	-0.327*** (0.075)	-0.515*** (0.053)	-0.030*** (0.010)	-0.049*** (0.014)	-0.075*** (0.025)
Peripheral Route (ELM)	Source Official Identity Salience	0.152** (0.064)	0.150* (0.079)	0.237*** (0.072)	0.038* (0.019)	0.048* (0.027)	0.099** (0.045)
	Scene Official Context Salience	-0.190*** (0.057)	-0.222*** (0.056)	-0.343*** (0.060)	-0.049** (0.021)	-0.091*** (0.030)	-0.067 (0.055)
	Engaging Narrative Style	0.105 (0.078)	0.107 (0.070)	0.258*** (0.078)	0.045* (0.023)	0.059** (0.029)	0.089** (0.039)
	Editing Frequency	0.104 (0.067)	0.136** (0.066)	0.277*** (0.076)	0.025 (0.019)	0.019 (0.028)	0.099** (0.049)
	Call to Action	-0.140* (0.075)	-0.117 (0.073)	-0.286*** (0.074)	-0.076*** (0.017)	-0.101*** (0.025)	-0.131*** (0.043)
	Product Visual Salience	-0.170*** (0.065)	-0.245*** (0.061)	-0.318*** (0.064)	-0.055** (0.021)	-0.076** (0.030)	-0.127** (0.053)
	Background Music	0.256*** (0.051)	0.253*** (0.052)	0.258*** (0.057)	0.092*** (0.029)	0.146*** (0.041)	0.169** (0.083)
	Speech Rate	-0.445*** (0.075)	-0.572*** (0.073)	-0.542*** (0.072)	-0.048*** (0.011)	-0.092*** (0.016)	-0.134*** (0.028)
	Linguistic Expression Accuracy	-0.050 (0.058)	0.114* (0.069)	-0.037 (0.053)	-0.025 (0.027)	-0.020 (0.040)	-0.045 (0.075)
	Video Duration	-0.072 (0.072)	-0.055 (0.079)	0.070* (0.041)	0.014 (0.010)	0.034** (0.014)	0.041 (0.025)
Control Variables	Follower Count	0.986*** (0.048)	0.570*** (0.049)	0.363*** (0.052)	0.052*** (0.010)	0.055*** (0.014)	0.093*** (0.024)
	Publisher Age	-0.008 (0.066)	-0.050 (0.069)	-0.112* (0.061)	-0.001 (0.010)	-0.003 (0.013)	-0.048* (0.024)
	Publisher Gender	-0.101 (0.061)	-0.125** (0.056)	-0.353*** (0.063)	0.012 (0.020)	-0.015 (0.029)	-0.125** (0.049)
	Publisher Province	0.004 (0.068)	-0.107 (0.076)	-0.188*** (0.064)	-0.011 (0.010)	-0.023 (0.015)	-0.075*** (0.025)
	Intercept	7.914*** (0.066)	5.836*** (0.066)	4.842*** (0.070)	1.830*** (0.042)	1.451*** (0.059)	1.207*** (0.113)
Model Fit	Pseudo-R ²	0.391	0.406	0.409	0.248	0.243	0.217

Table A5

Propensity Score Matching (PSM)–Based Estimates of the Average Treatment Effect on the Treated (ATT).

Strategy Variable	Likes (ATT)	Comments (ATT)	Shares (ATT)
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Table A5 (continued)

Strategy Variable	Likes (ATT)	Comments (ATT)	Shares (ATT)
Content_Information_Complexity	0.272**	0.257**	0.416***
Marketing_Information_Intensity	-0.265*	0.049	0.243***
Content_Rational_Appeal_Strength	-1.119***	-1.266***	-1.174***
Source_Official_Identity_Salience	0.118***	0.116**	0.365***
Scene_Official_Context_Salience	-1.062***	-1.108***	-0.727***
Engaging_Narrative_Style	0.652*	0.670**	0.688***
Editing_Frequency	0.566***	0.643***	0.914***
Call_to_Action	-0.908***	-0.785*	-1.037***
Product_Visual_Salience	-0.771***	-1.506***	-0.414***
Background_Music	0.754*	1.553***	0.752*

Table A6

Analytical Transparency and Alternative Validation Logic.

Analytical step	Primary choice in this study	Alternative or baseline considered	Validation logic and boundary
Multimodal feature extraction	Sequential context-injection with dynamic contrastive analysis comparing frame t with frame $t-1$.	Static or independent frame coding.	Chosen to preserve temporal cue changes in short videos. This design improves analytical fit for dynamic media but should not be interpreted as a causal identification strategy.
Topic discovery	BERTopic applied to MM-LLM-generated semantic annotations.	LDA-style bag-of-words topic modeling.	Chosen because the annotations are short-text-like, semantically dense, and lexically diverse. This is a methodological fit argument; no formal numerical claim is made that BERTopic outperforms all topic models.
Topic-to-construct synthesis	C_v topic coherence diagnostics, topic-size filtering, qualitative pruning, HAC with Ward's linkage, and researcher review.	Accepting raw topic labels or relying only on coherence scores.	The 85 topics $\rightarrow$ 47 validated topics $\rightarrow$ nine constructs chain combines quantitative diagnostics with semantic interpretability, representative examples, c-TF-IDF inspection, and preparatory review before formal theory anchoring.
Scaled coding reliability	Two independent AI coding instances plus human adjudication for divergent or context-dependent cases.	Single AI run, unadjudicated AI output, or simple averaging of AI outputs.	AI-AI, human-human, and AI-human diagnostics identify where AI is reliable and where human judgment is necessary; final adjudication is evidence-based and rule-guided.
Model-specific stability	Gemini 2.5 Pro as the main inference model, with Gemini 2.5 Flash validation on a stratified subsample.	Reliance on a single foundation model.	Cross-model validation on $N = 78$ shows strong convergence with the final dataset, supporting procedural stability across two Gemini variants while not claiming universal model independence.

Table A7

Full-Sample Interaction-Based Coefficient-Difference Tests for Heterogeneity Analysis.

Variables	Video Long - Video Short	Fans Large - Fans Small	Gov Good - Gov Poor
Content Information Complexity	0.412*** (0.145)	-0.494* (0.280)	-0.857*** (0.291)
Marketing Information Intensity	-0.508*** (0.161)	0.549 (0.352)	0.391** (0.165)
Content Rational Appeal Strength	-0.190 (0.441)	-0.055 (0.636)	0.739 (0.503)
Source Official Identity Salience	0.421** (0.207)	-0.828*** (0.223)	-0.752*** (0.230)
Scene Official Context Salience	0.600* (0.344)	0.227 (0.448)	-0.343** (0.145)
Engaging Narrative Style	-0.607* (0.347)	-0.103 (0.474)	0.174 (0.476)
Editing Frequency	0.630* (0.362)	-0.603** (0.291)	-0.375 (0.572)
Call to Action	-0.551* (0.314)	0.033 (0.548)	-0.361 (0.497)
Product Visual Salience	-0.214 (0.462)	0.242 (0.472)	0.300 (0.409)
Background Music	0.128 (0.233)	0.388 (0.377)	0.385 (0.377)
Speech Rate	-0.215 (0.399)	-0.586 (0.479)	-0.611 (0.495)

Note: Cells report interaction coefficients from full-sample negative binomial models predicting shares. Column labels indicate contrast direction. Standard errors are in parentheses; controls are included. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

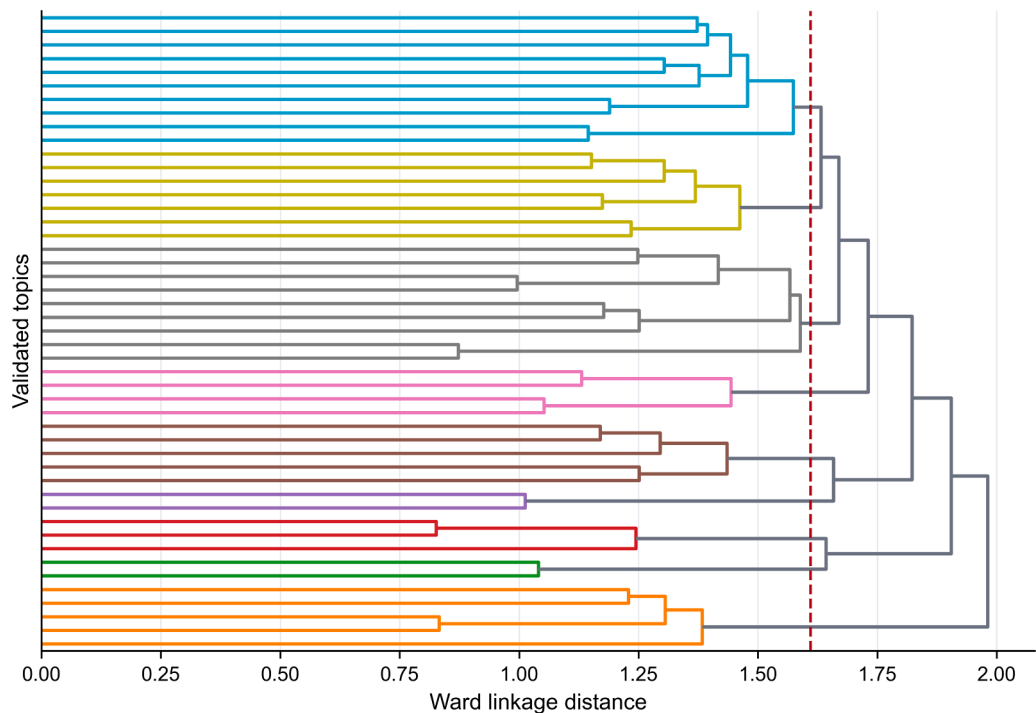


Fig. A1. HAC dendrogram for the 47 validated topics. Note. The dendrogram was generated using Ward’s linkage with Euclidean distance on TF-IDF topic-document vectors. The dashed vertical line indicates the tree cut used to obtain nine candidate clusters. Colors denote HAC candidate clusters prior to ELM-based theory anchoring and construct-label auditing.

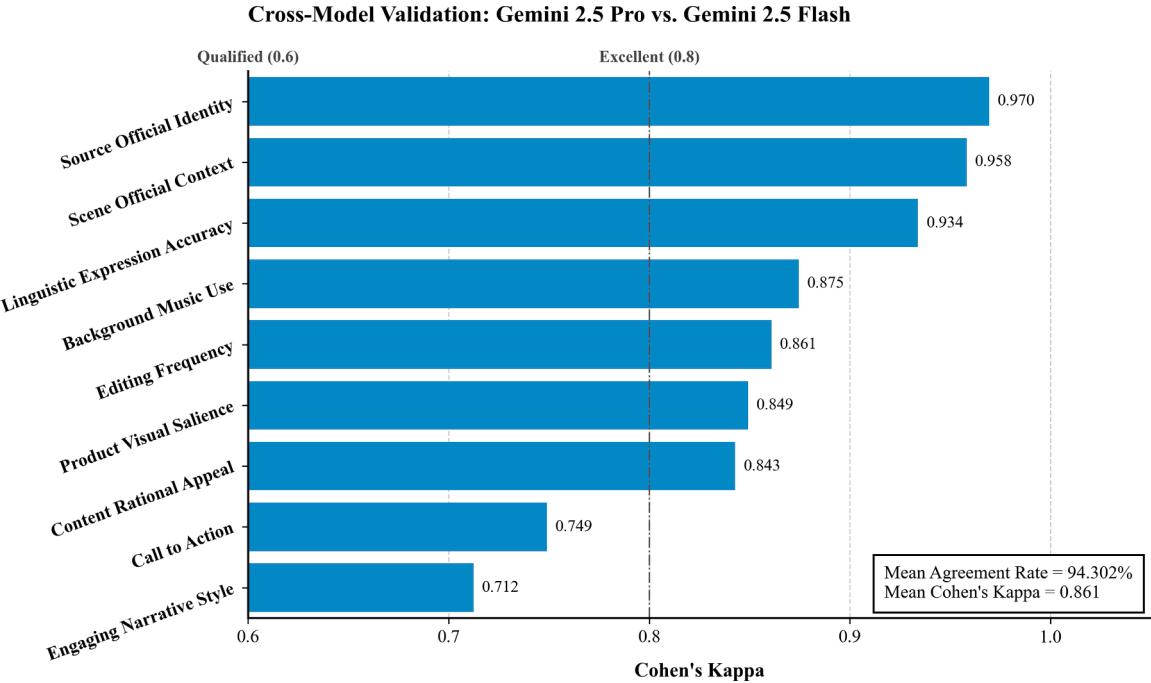


Fig. A2. Cross-Model Validation Metrics.

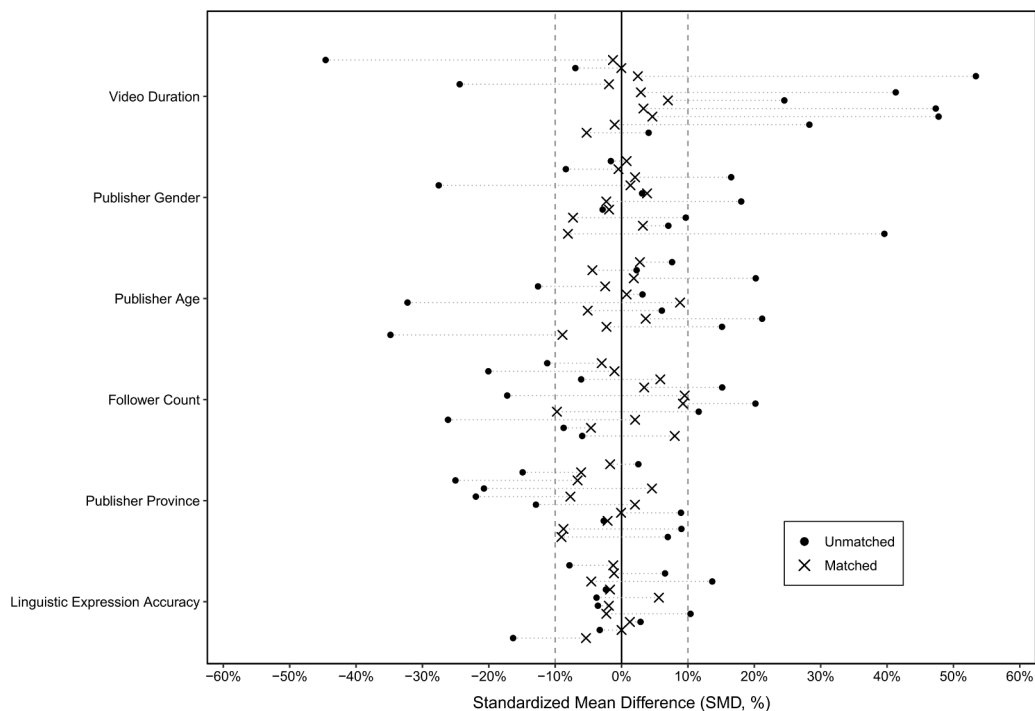


Fig. A3. Treatment-Specific Covariate Balance Diagnostics After Propensity Score Matching. Note: Solid circles indicate unmatched SMDs and crosses indicate matched SMDs for treatment-specific covariate balance. Dashed vertical lines mark the conventional $\pm 10\%$ threshold.

¹**Cost Analysis:** The total cost for the double-blind analysis of 779 videos (1,558 requests) was approximately \$43.86. This includes 10.13 million input tokens (video/audio/text context at \$1.25/1M) and 3.12 million output tokens (structured coding data and Chain-of-Thought at \$10.00/1M).

Data availability

The data that support the findings of this study are available from the author upon reasonable request.

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